أكاديمية الفضاء الوطنية
NATIONAL SPACE ACADEMY

وكالة الإمارات للفضاء
UAE SPACE AGENCY

Intro to hyperspectral remote sensing

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01

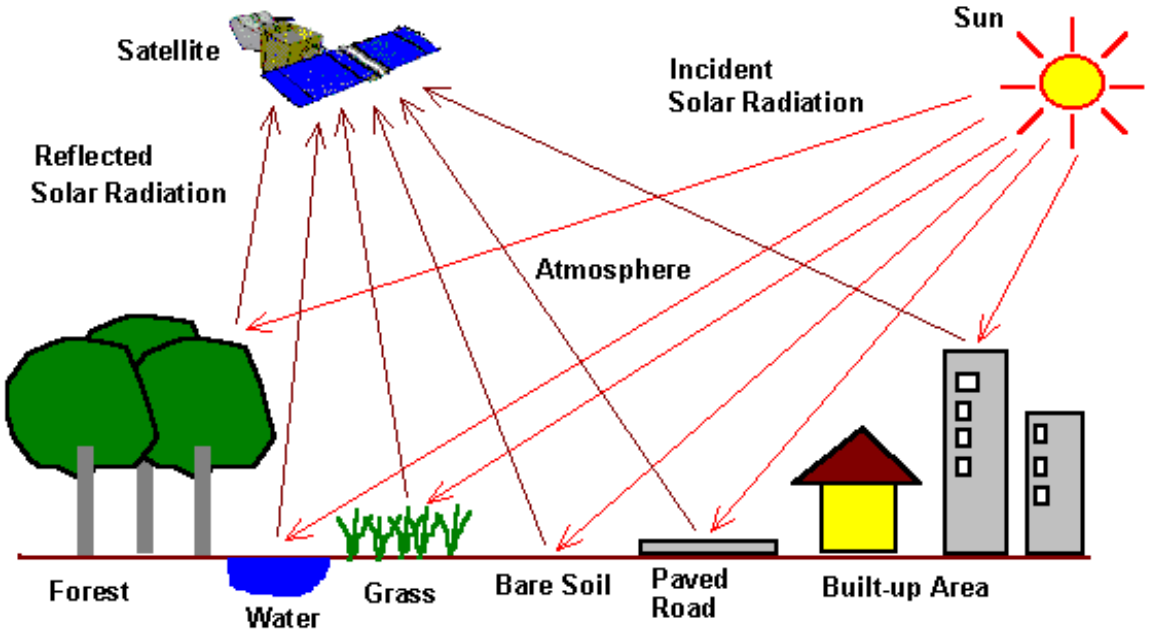
Spectroscopy



Recap: intro to remote sensing

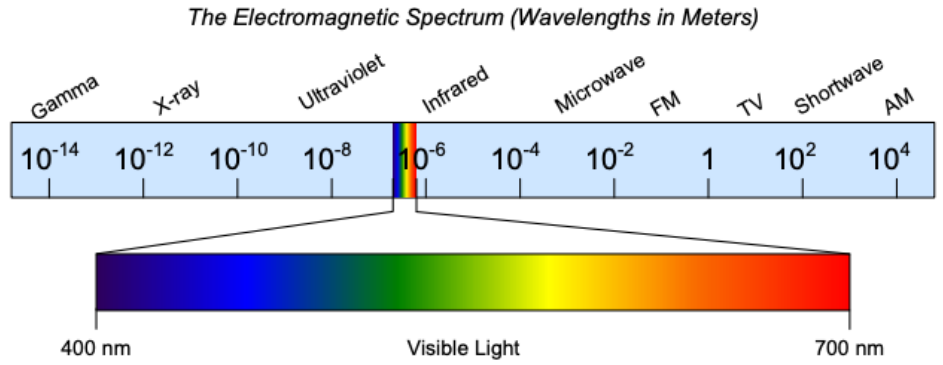
- RS: the science of gathering data from a distance
  - Requires a source of illumination
- Each object has its own spectral signature: proportion of incoming vs reflected energy
- 2 main types:
  - Active: uses an active signal, laser (LiDAR) or Radar
  - Passive: optical RS uses the sun
- Methods of acquisition:
  - Satellite, airplane, drone
- The objective determines the required data characteristics
  - Forest change detection: low spatial and low temporal resolution
  - Crop health monitoring: high spatial and temporal resolution
  - Global land cover change

- RS is applied in almost any industry
  - Forestry, agriculture, defence, weather, ...
  - Applications are endless.



Spectroscopy & EMR

- Study of wavelength composition of electro magnetic radiation
- Electro magnetic radiation (EMR) is energy in electro magnetic waves
  - Visible light, microwave, xray
- EMR can be transmitted, absorbed, or refracted



Refraction

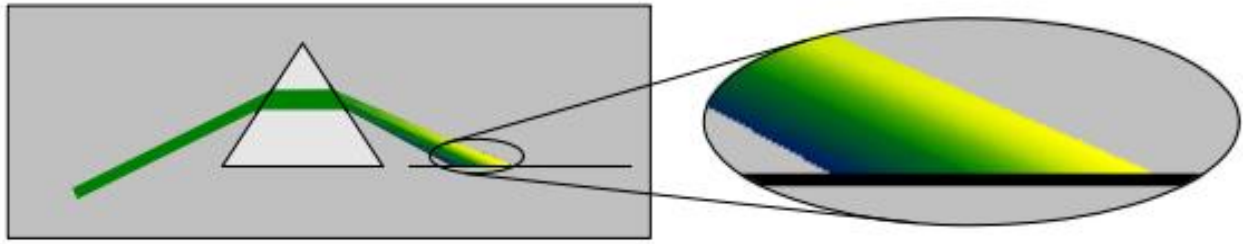
- When EMR passes through a substance, waves change direction/bend

Visible light comprises of many bands

- White light is a mix of all colors

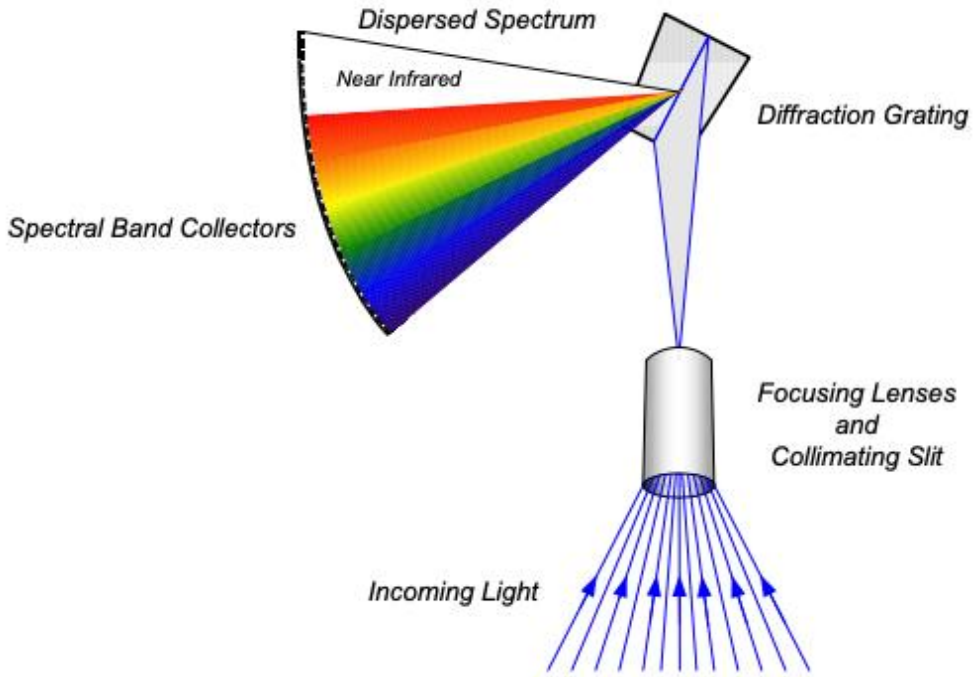
Natural white light is a mixture of wavelenghts

- E.g. green light from a tree is a mixture of blue, green and yellow light



How does a spectroscopy sensor work?

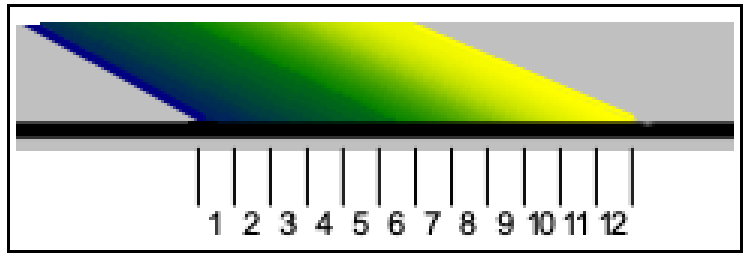
- A scan mirror with lens and slit redirects the incoming light
- A dispersing element refracts the light into separate spectral band collectors
- Diffraction grating are etched at specific angles/widths and spacing, refracting the light into channels (1000s per mm)



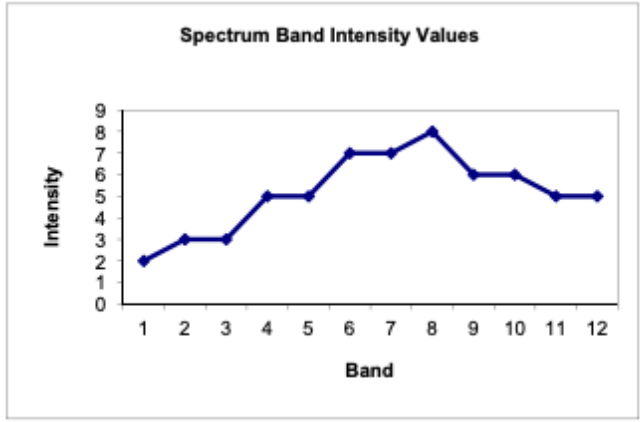
Measuring the spectrum

- Infinite number of colors & intensities in visible spectrum
  - Each unique color has a different intensity
- Using refraction we can disperse unique spectra and intensities into bands

- The separate bands have unique light values
  - The green light intensity is measured
  - Data is assigned an arbitrary values from 1-10



Band	Intensity Value	Band	Intensity Value
1	2	7	7
2	3	8	8
3	3	9	6
4	5	10	6
5	5	11	5
6	7	12	5



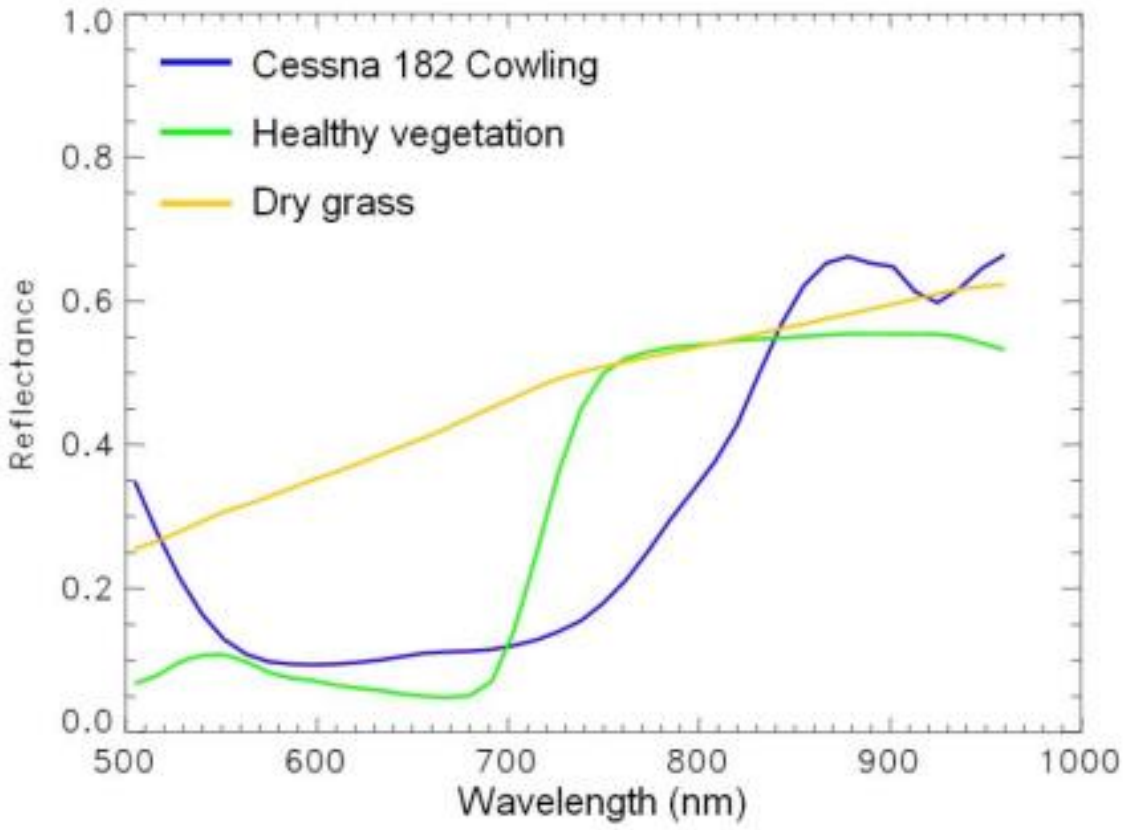
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Spectral signatures

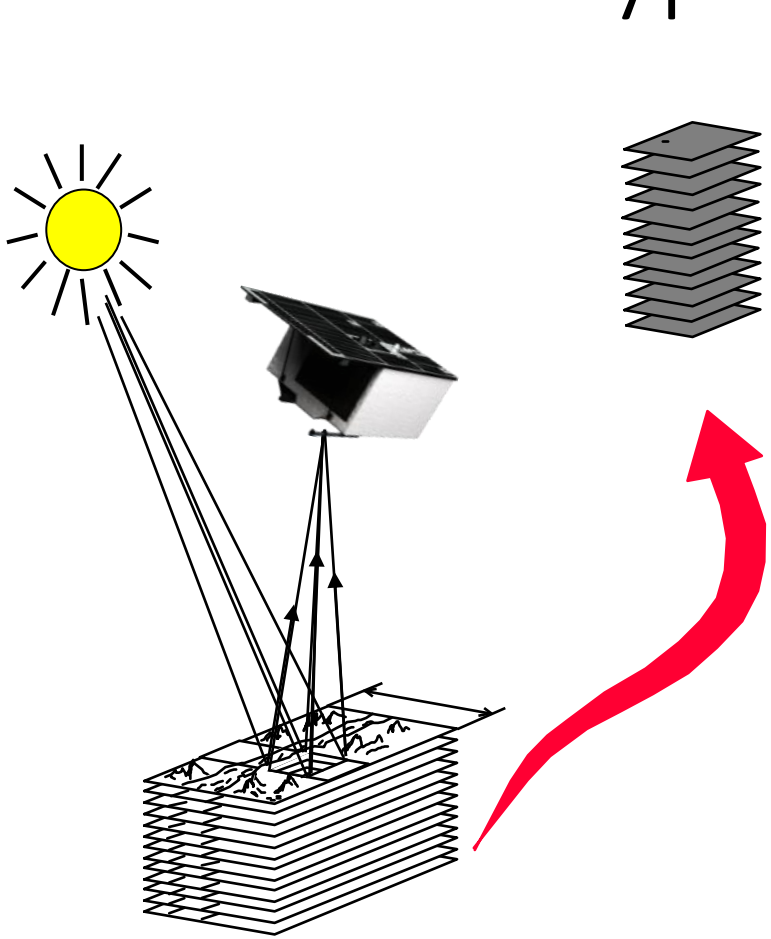


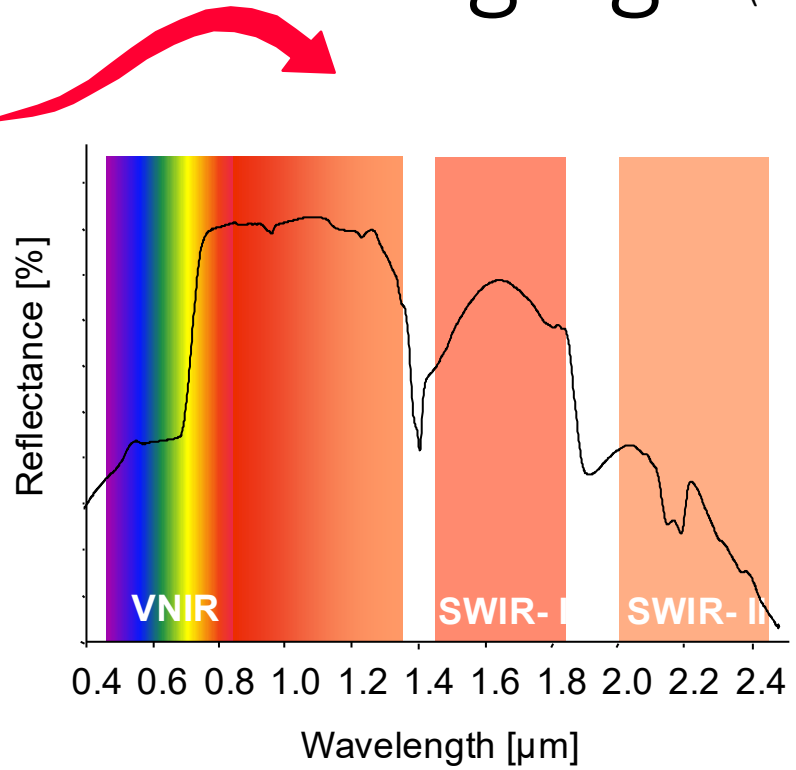
Spectral signature

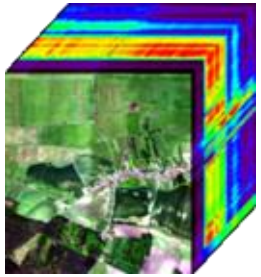
- Using light refraction we can collect unique intensity signatures
- The refracted light can be used to discriminate similar looking objects
- Reflectance is a physical property
  - % of reflected incident EMR per wavelength
- Unique fingerprint
- Spectral signature depends on:
  - The source that illuminates the object
  - The reflectance of the object



What is Hyperspectral imaging? (HSI)





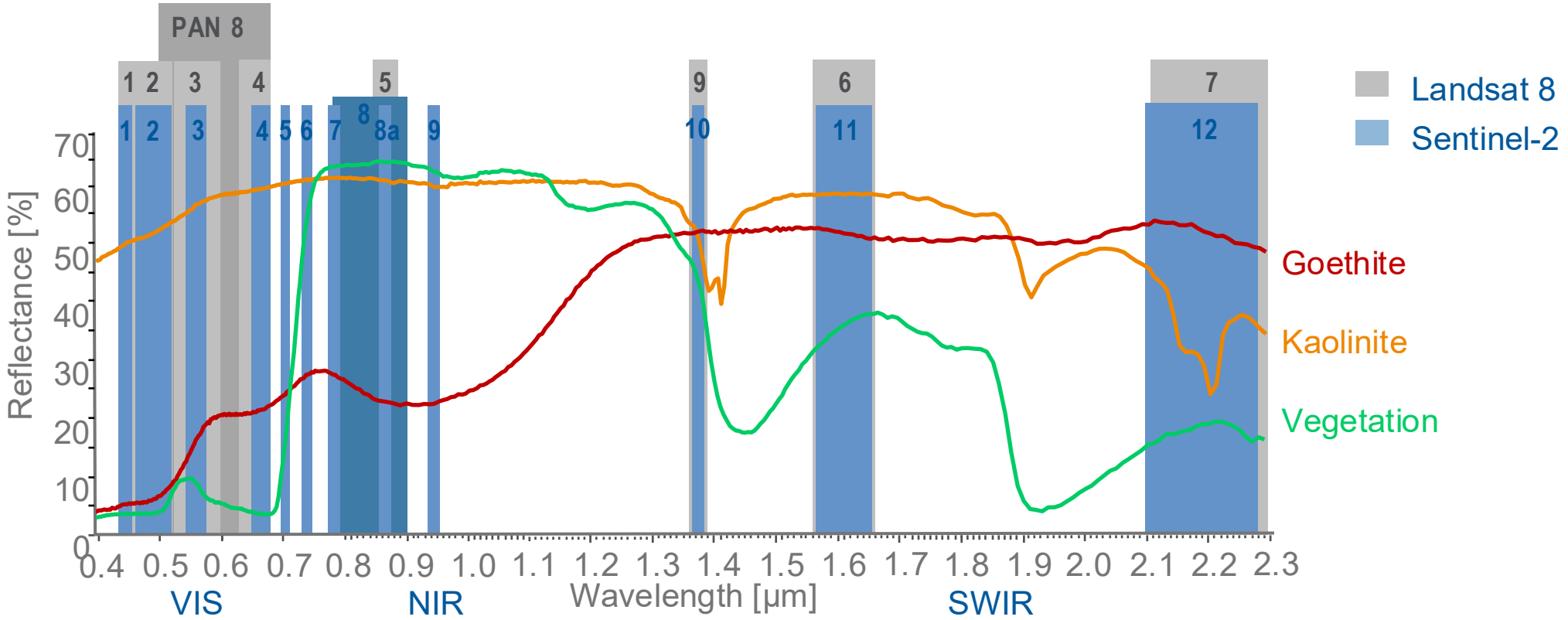


- Collect (image) data in many narrow contiguous spectral bands.
- Creates a complete record of spectral responses of the material
- 2 main instrument types:
    1) Image data of many rows of pixels
    2) Spectral data of a measured spot

Multi vs hyperspectral

Multispectral: Usually 4 – 12 spectral bands
Hyperspectral Usually ~100s of spectral bands
More spectral bands = better material separability

Many surface information can be derived **more accurately** with hyperspectral data, while they are **essential** for certain applications



Multi vs hyperspectral

Multispectral:

– Popular sensors such as Landsat and MODIS
– Limited number of spectral bands
– Relatively high temporal resolution
– Global spatial extent

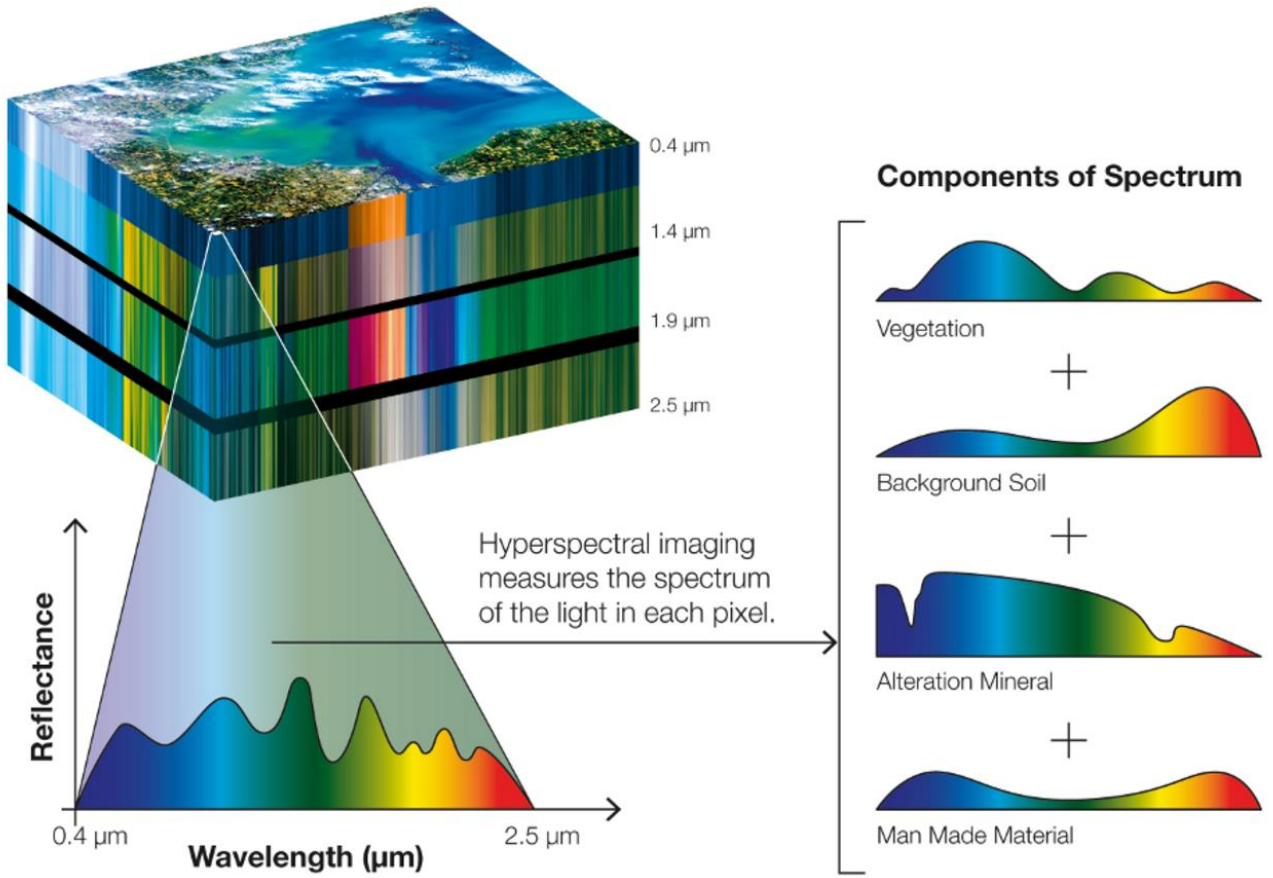
Hyperspectral

- Limited in numbers of satellite sensors
– Some mission-specific sensors aboard the ISS
– Airborne sensors flown during flight campaigns
– Narrow bands that measure more characteristics of surface reflectance
– Low temporal resolution
– Less spatial coverage

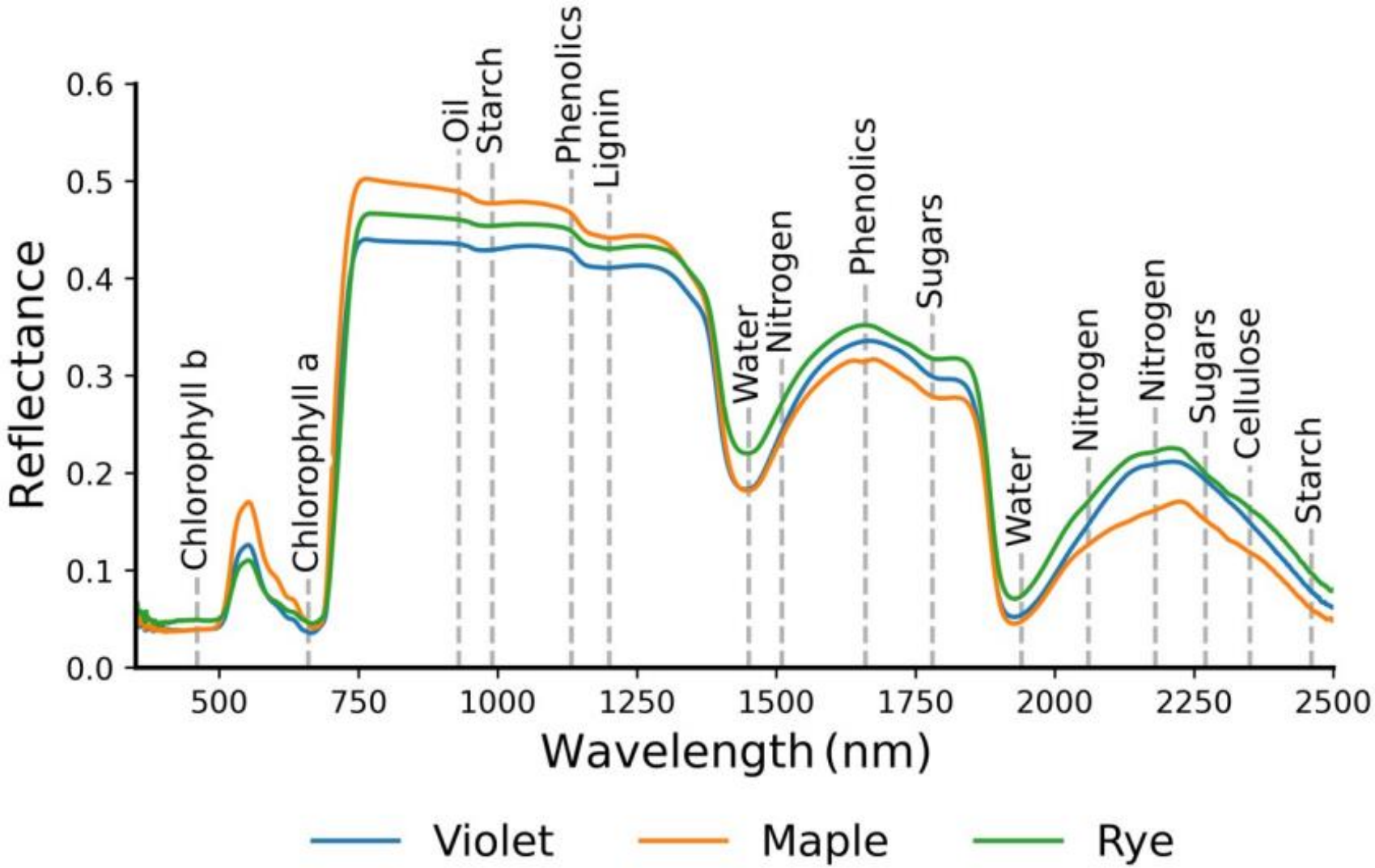
Spectral responses

- Powerful inspection method:
  - Reliable identification of similarly looking but different chemical characteristics
  - Quickly growing in computer vision industry
  - Camera costs decreasing
  - High commercial value

- Objects have their own spectral signature
  - Includes atmospheric effects due to water vapor, gas constituents, aerosols, etc.



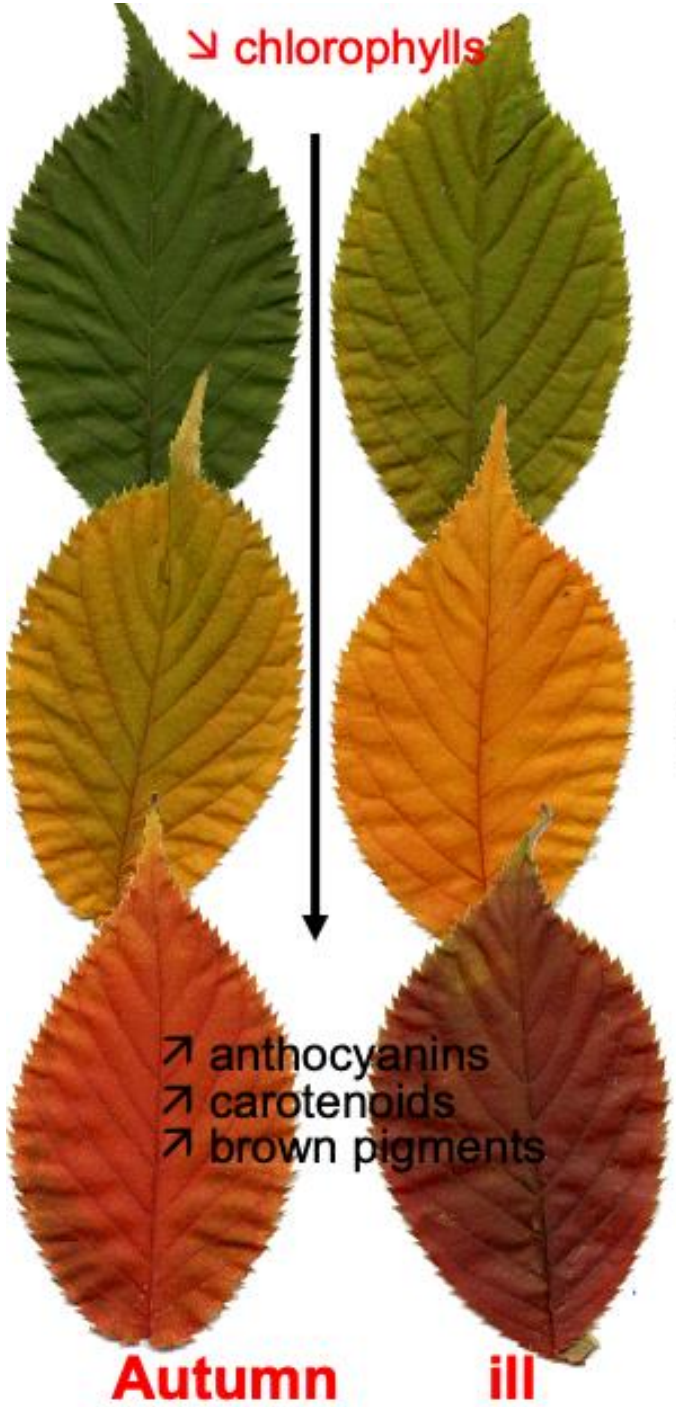
Reflectance Spectroscopy



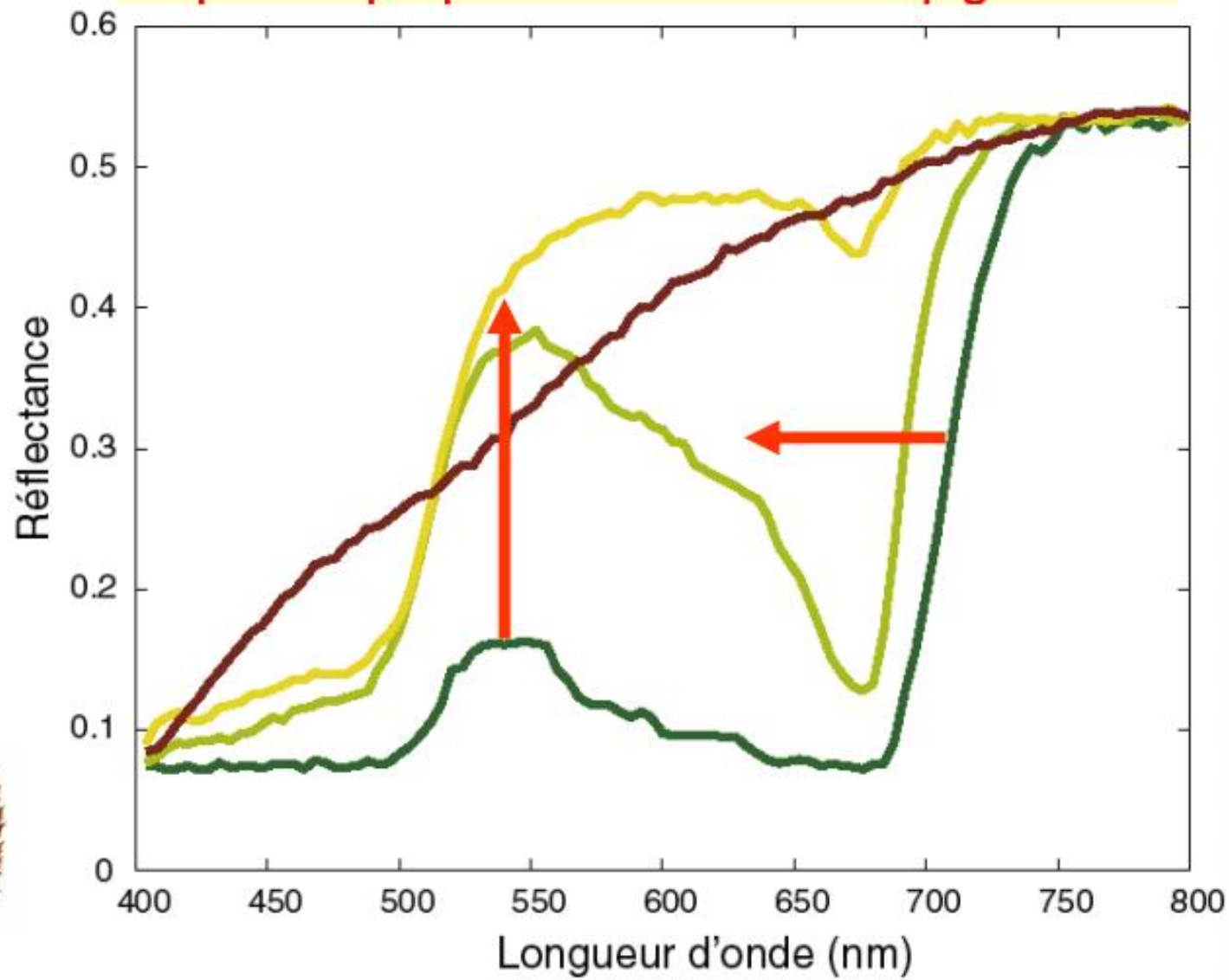








Spectral properties: effect of leaf pigments



3

Data acquisition methods



Data acquisition

Field spectrometers

- Used to collect field in-situ data for validation purposes
  - Assessing material composition
  - Identifying substances
  - Collecting spectral databases
- Used in various industries
  - Agriculture, forestry, geology,
- High spectral resolution
  - 400-2500nm
- Light weight, Reliable and portable



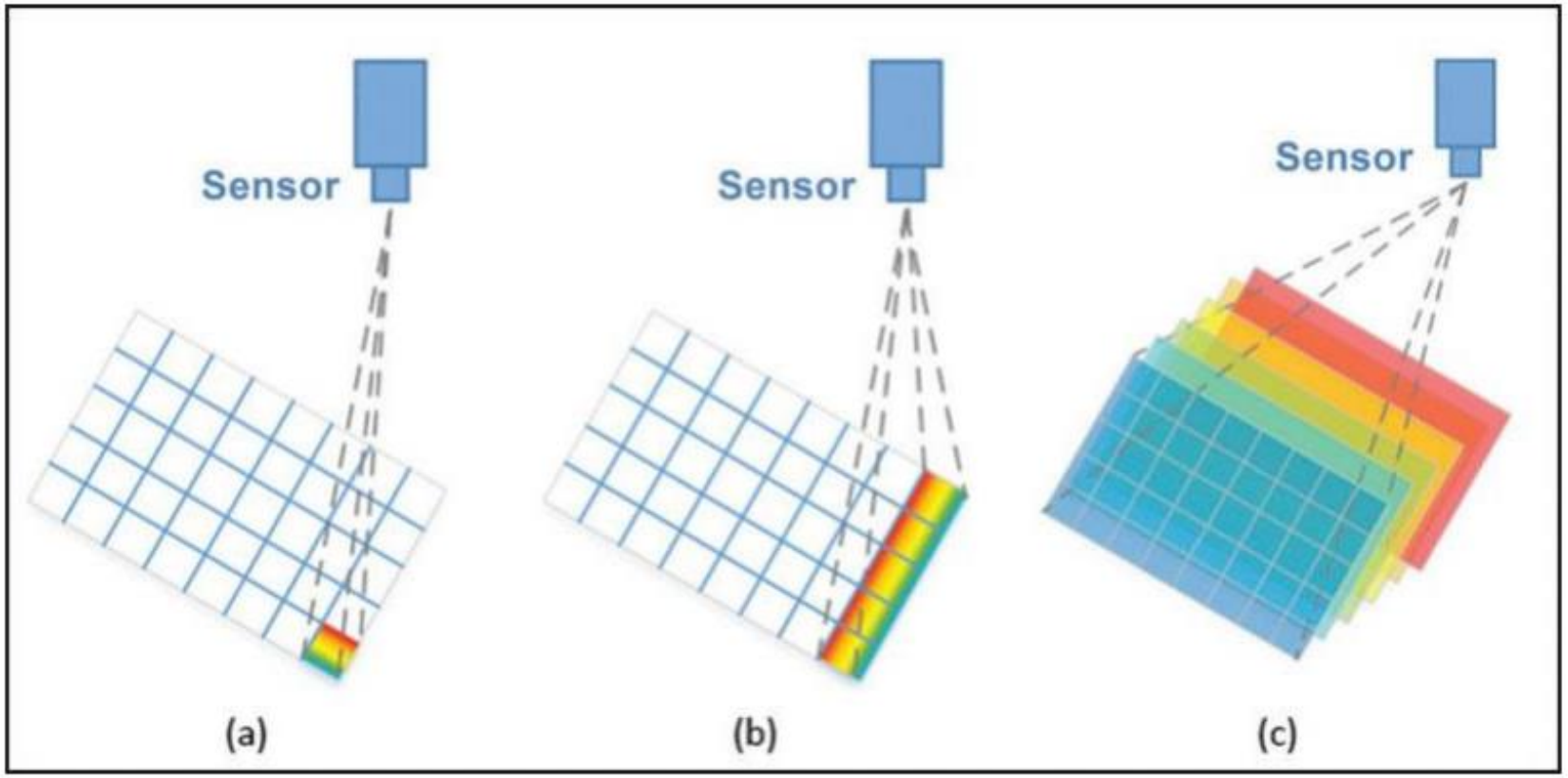
Data acquisition (2)

- Airplane
  - Reliable
  - Steady
  - Covers large area fast
- Drone/UAV
  - Low acquisition cost
  - Good for small area
  - Challenging with heavy hyperspectral sensors
  - Commonly used with smaller/lighter multispectral cameras





Image acquisition modes



whiskbroom pushbroom Frame camera

Main principle of hyperspectral imaging

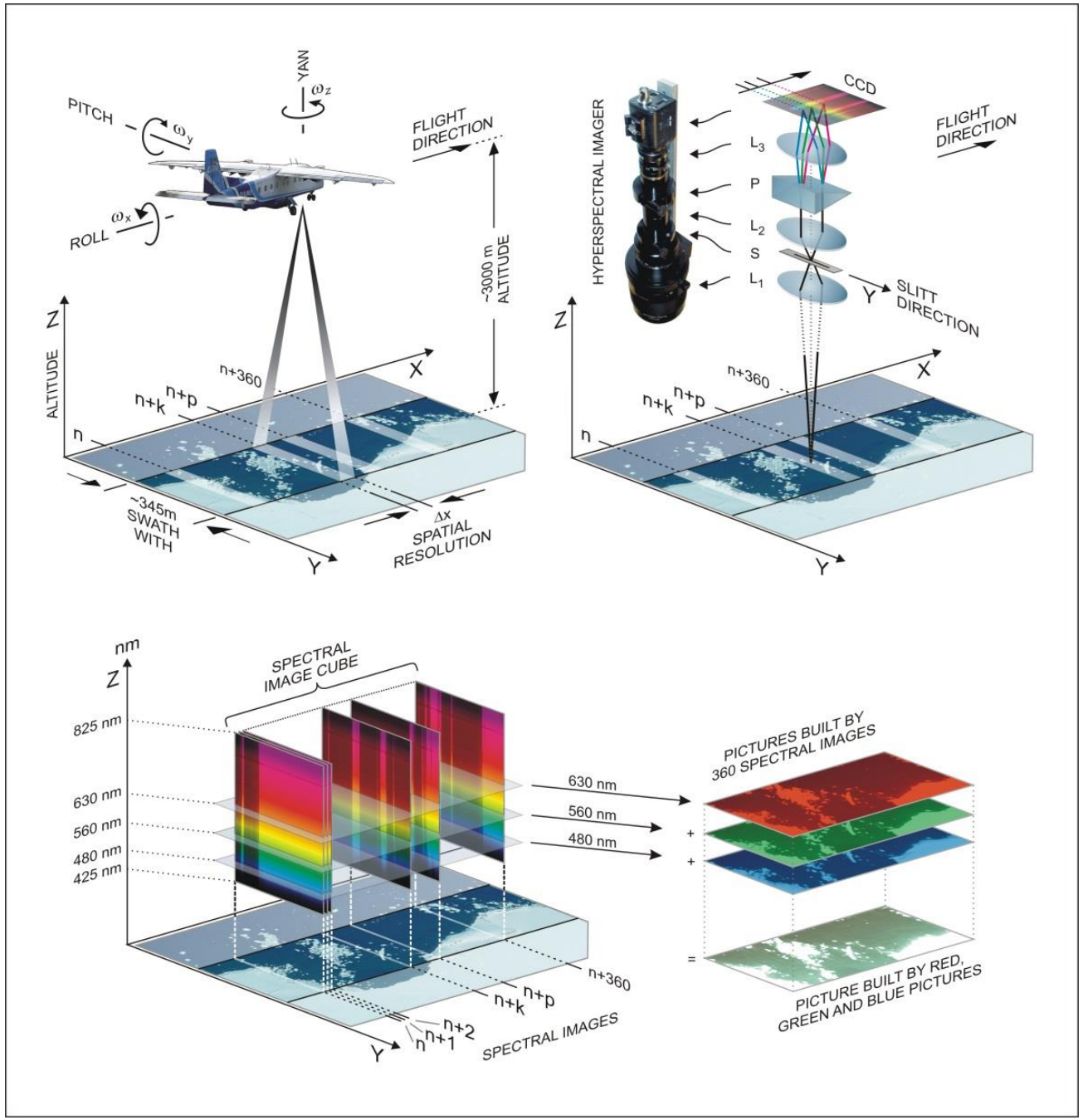


Image technique

In order to sample the whole Target object, we need to rotate the instrument or fly over it.

Or the slit entrance image needs to sweep across the entrance slit plane (rotating mirror).

This is known as the **pushbroom** technique.

The net result is a spectral cube or
spectral movie.

We can now generate images as a function of wavelength.

（1）Whiskbroom Imaging Spectrometer

Image technique

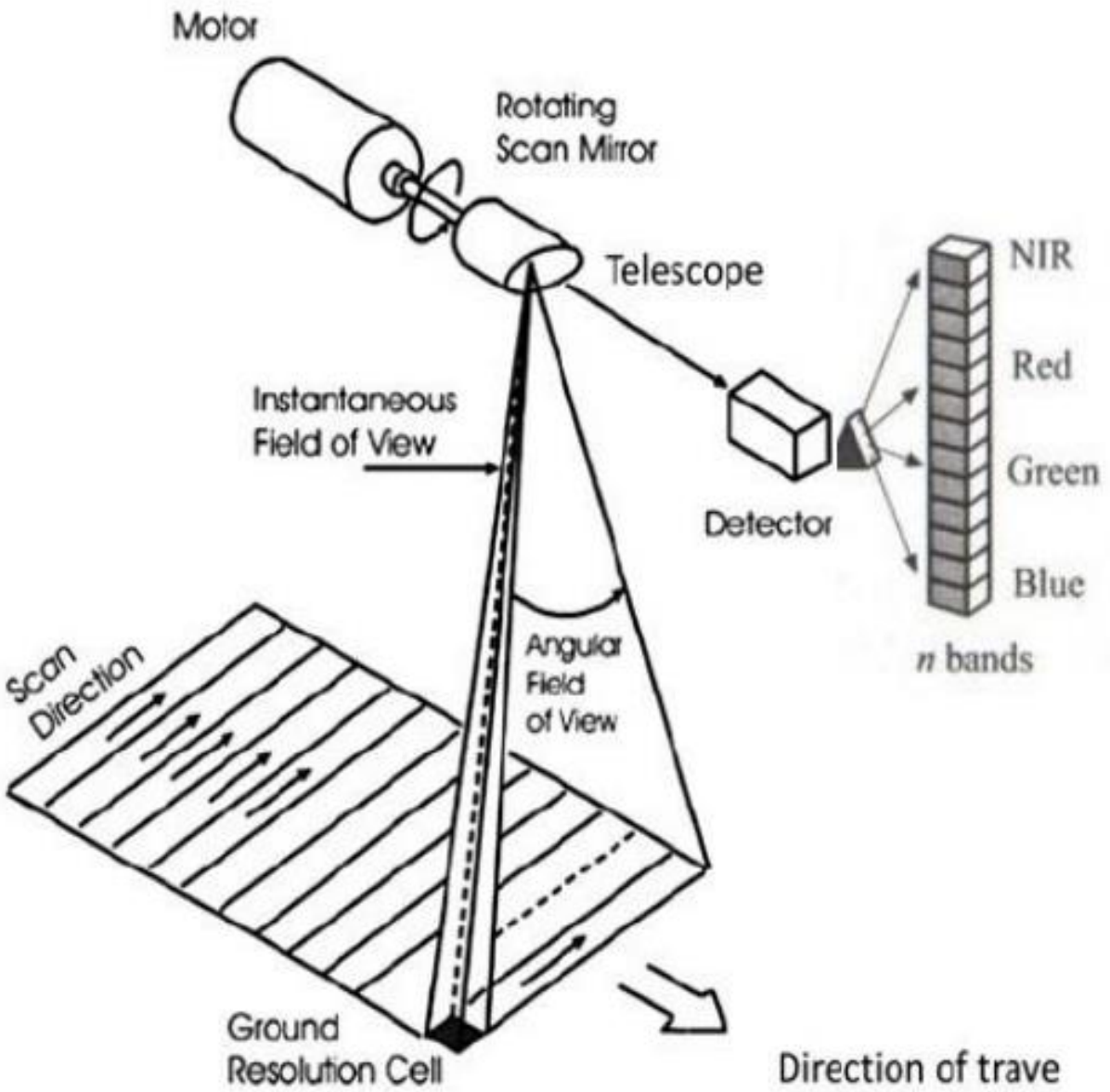
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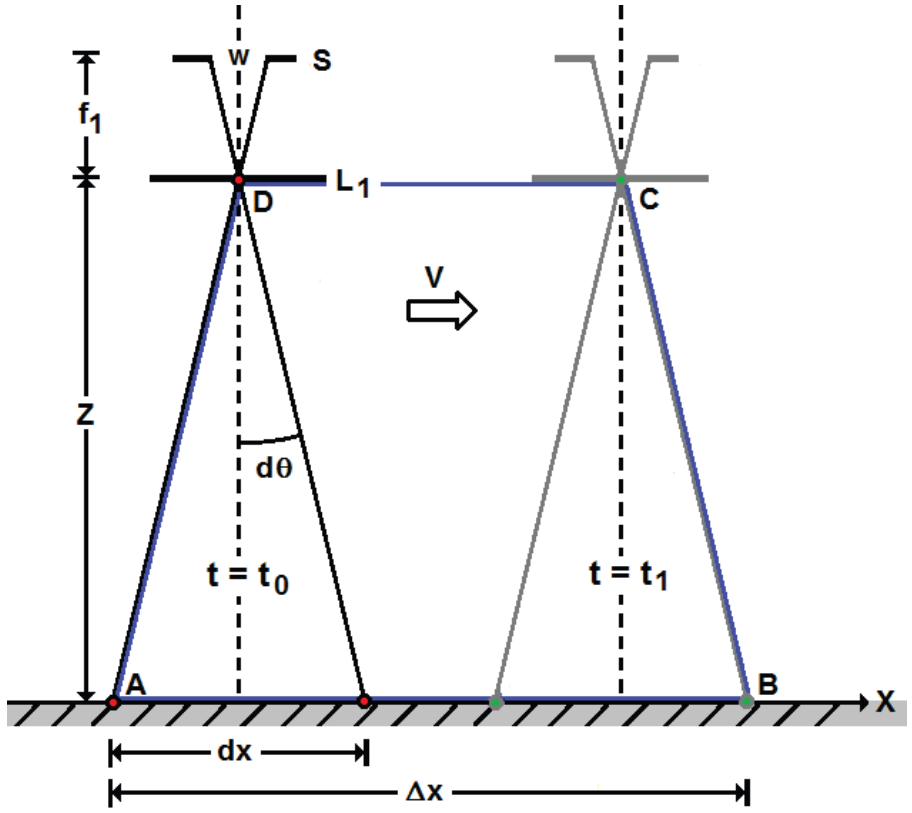
This is known as the **pushbroom** technique.

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spectral movie.

We can now generate images as a function of wavelength.



Hyperspectral image formation and spatial resolution



Our instruments:
w = 0.025 mm
h = 3 mm
f_1 = 16 mm

Note that

$$v \cdot \tau \le dx$$

$$d\theta = \arctan(\frac{w}{2f_1}) = 0.045^o$$

$$\Omega = 2 \times \arctan(\frac{h}{2f_1}) = 10.7^o$$

$$SW = 2z \times \tan(\frac{\Omega}{2})$$

DC: $v \cdot (t_1 - t_0) = v \cdot \Delta t$

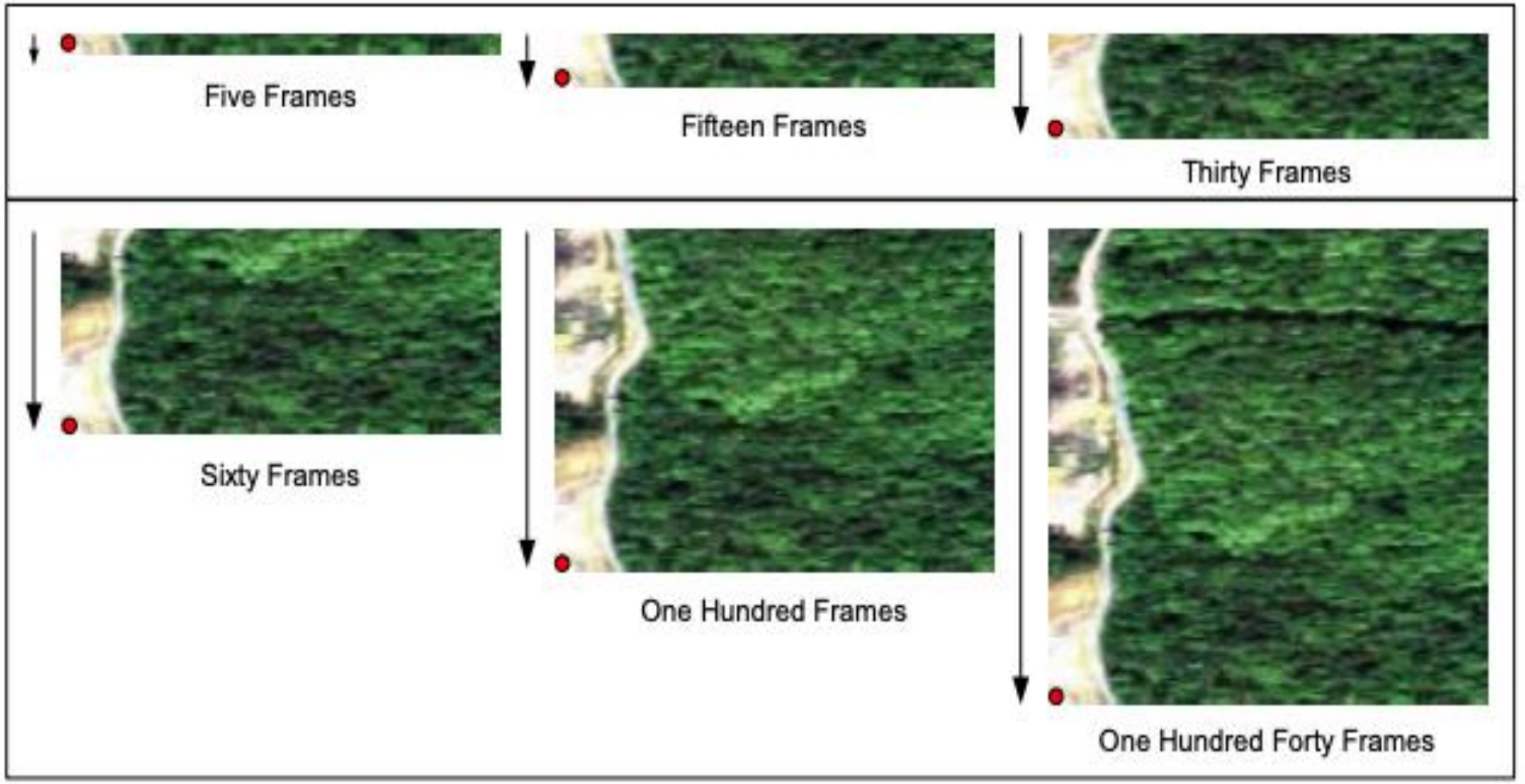
AB: $\Delta x = dx + v \cdot \Delta t$

$$dx = \frac{z \times w}{f_1}$$

$$\Delta y = \frac{z \times h}{f_1 \times N}$$

Z	dx	Δx	SW
100	0.16	0.36	18.75
300	0.47	0.67	56.25
500	0.78	0.98	93.75
1000	1.56	1.80	187.5

Table 1. Example calculations.
Parameters: Δt = 20 ms (1/50) s., 25 frames per second, read out time t = 20 ms and speed v = 10 m/s (36 km/t). All numbers in meters.



The framerate indicates output signal per second.

E.g.: At 60 HZ (1/60th) every 60th of a second a new row of pixels is captured.

- The image ‘grows’ as the plane moves forward.

This creates a waterfall pattern

link



04

Processing HSI data



Processing HSI data

- 1) Radiometric calibration
  - Convert raw digital numbers (DN) to a physical quantity (radiance)

  2) Spectral & spatial binning
  - - Reduce signal to noise ratio by combining spectral channels or neighbouring pixels.

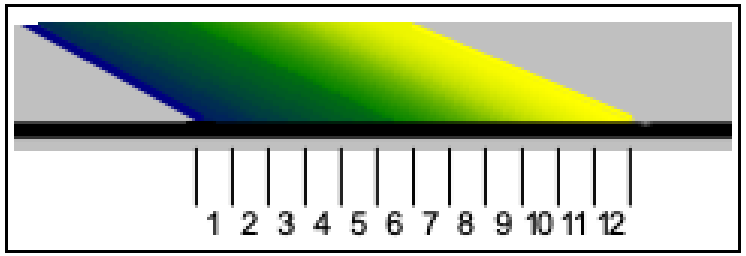
- 3) Atmospheric correction
  - Removal of atmospheric aerosols
  - Correct for lens variability with a dark/white reference
  - Results in surface reflectance

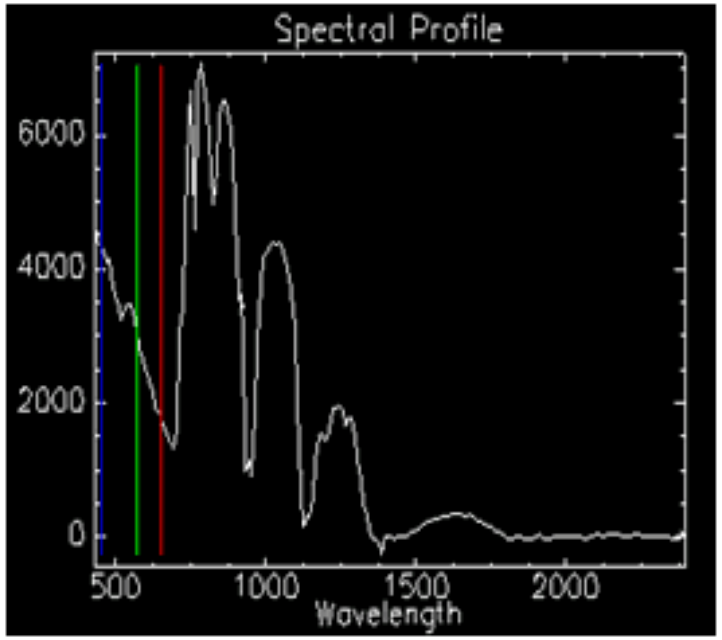
- 4) geometric correction
  - - Rectifying the image with control points (Lidar, or orthoimage)

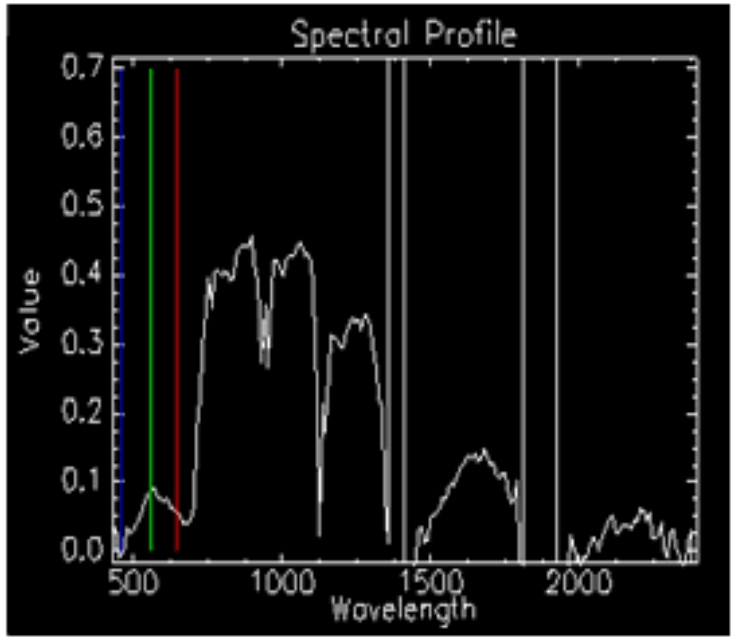


Radiometric calibration

- Assigns a wavelength to each channel
- Convert raw digital numbers (DN) to physical radiance values
- Dark current correction
- BRDF correction







Spectral & spatial binning

HSI data is large and complex to process

Two common techniques to reduce size and noise:

- Spectral binning
- Spatial binning

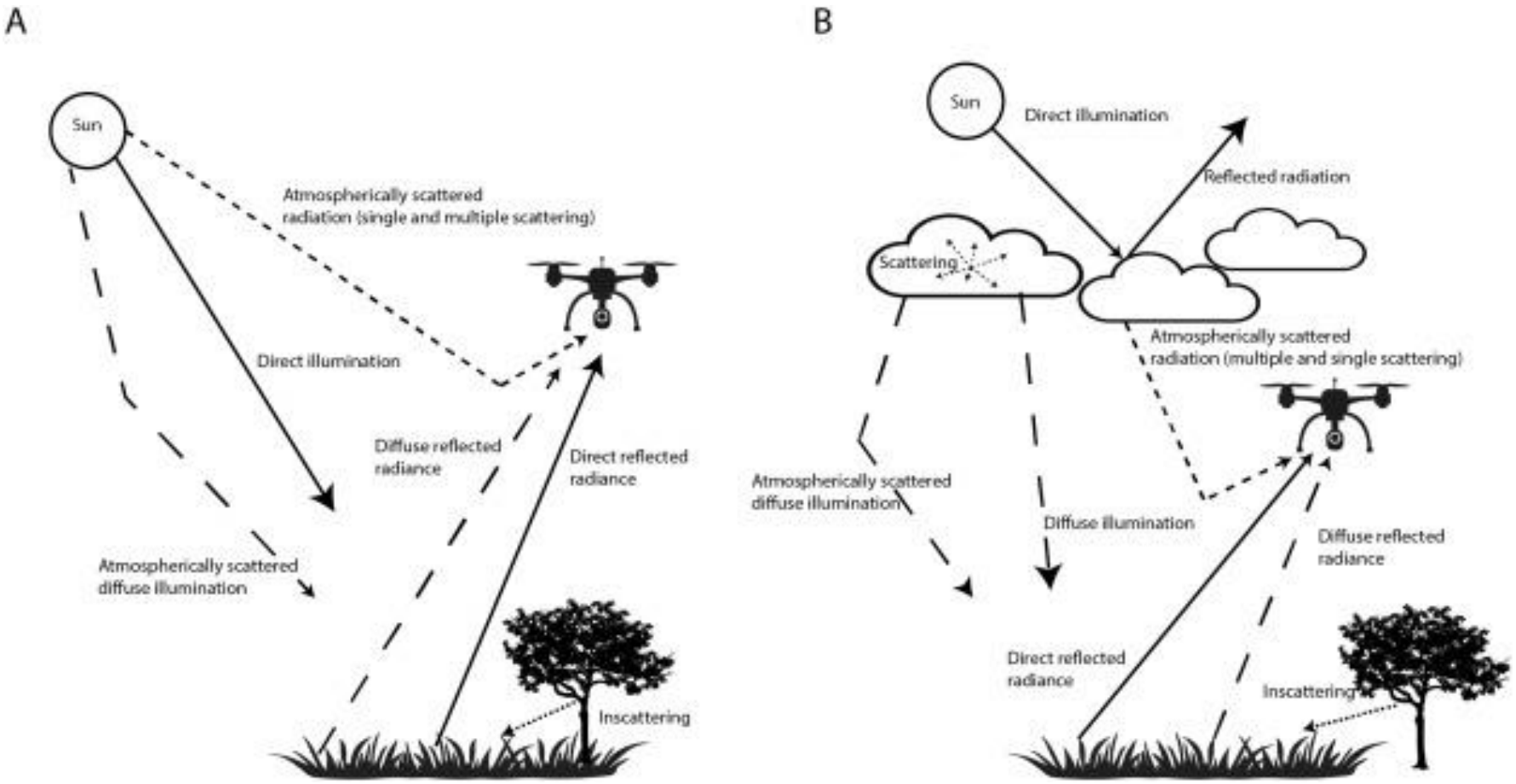
Spectral binning:

- Grouping spectral bands together
- Increases signal to noise ratio

Spatial binning:

- Grouping pixels to decrease spatial resolution
- Increases computation speed

Atmospheric correction(1)



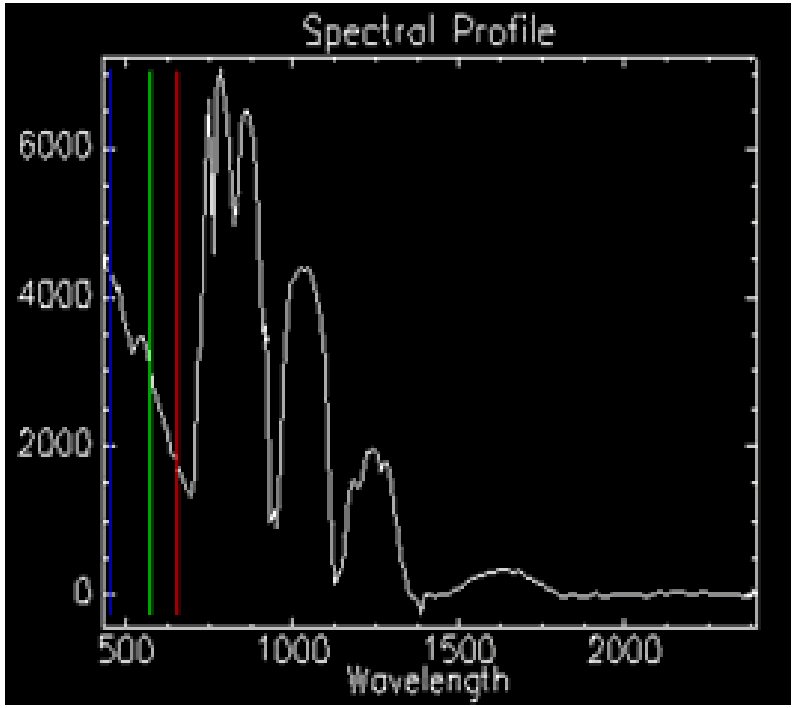
Radiation travels through atmosphere and interacts with vapors, aerosols and gasses.

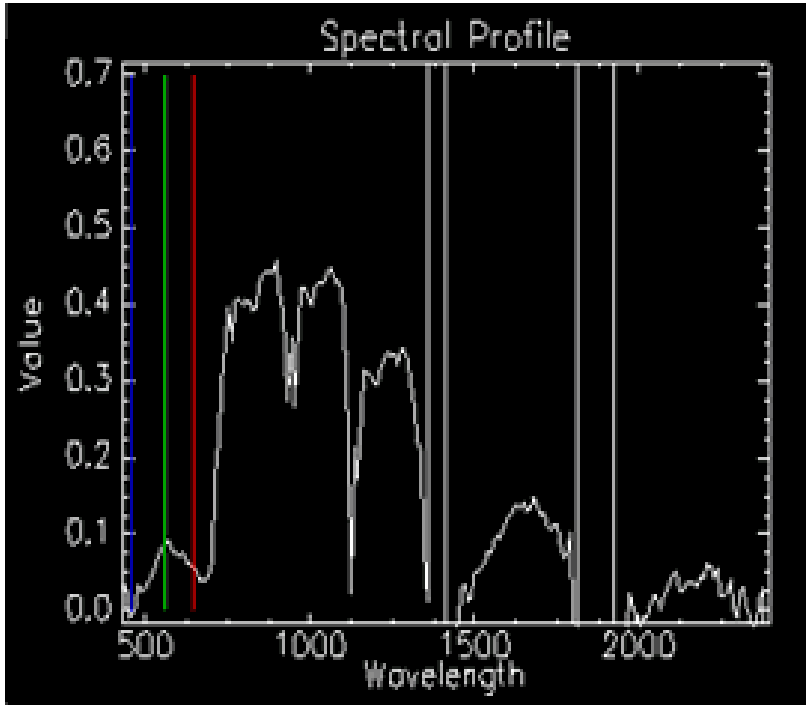
The EM radiation usually has 3 pathways:

- Direct transmission
- Scattered (single vs multi)
- Diffused

Most blue light is reflected by the atmosphere, clowds, and aerosols.

- We remove most radiation in the water absorption bands to get more accurate





Geometric correction

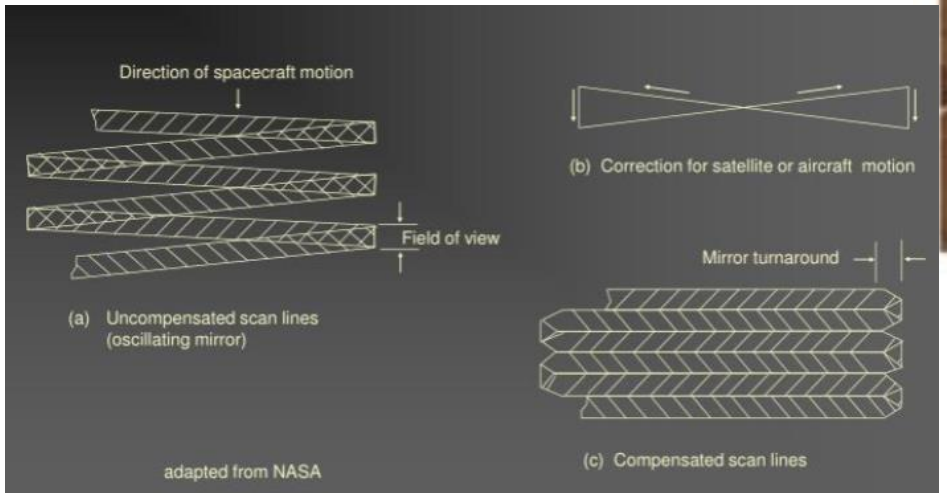
Orthorectification is converting images suitable for maps by removing:

- Sensor, satellite/aircraft motion, and terrain related geometric distortions

We need ortho datasets for thematic analysis

- Data fusion
- Overlaying images with datasets
- Evaluating change detection and updating maps

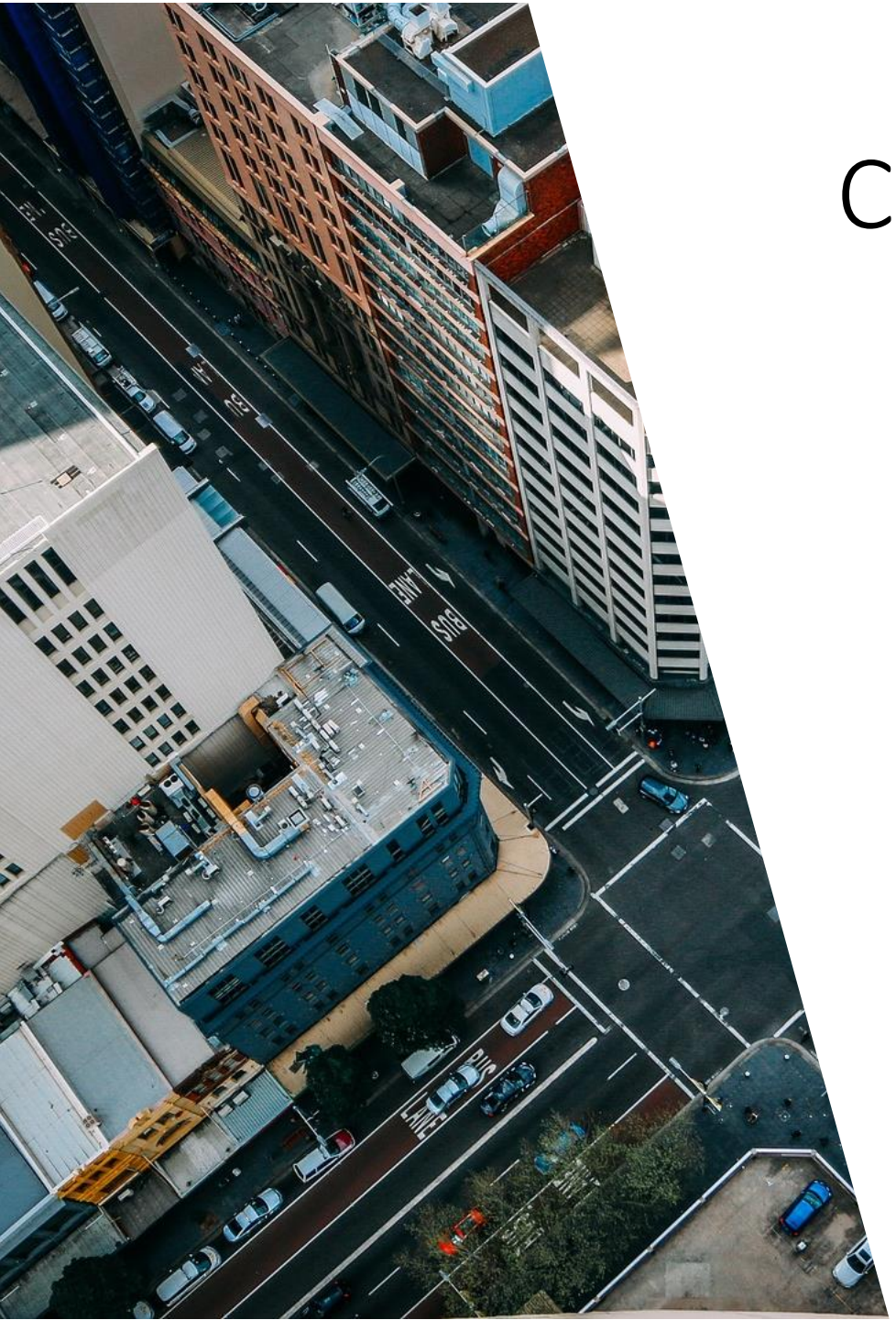




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Challenges





Challenges (1)

- Illumination conditions
  - HSI requires absencece of clouds
  - Sun should ideally be perpendicular to the sensor
  - Sun angle should be high
    - Low angle causes long shadows

- Atmospheric conditions
  - Haze, dust or smoke should be avoided
    - 2019 Forest fires above borneo, Indonesia with S2

Challenges – illumination conditions

- Illumination conditions and environmental conditions
  - Sun angle will affect how the target appears to the sensor
  - Haze and clouds cause more absorption and light scattering

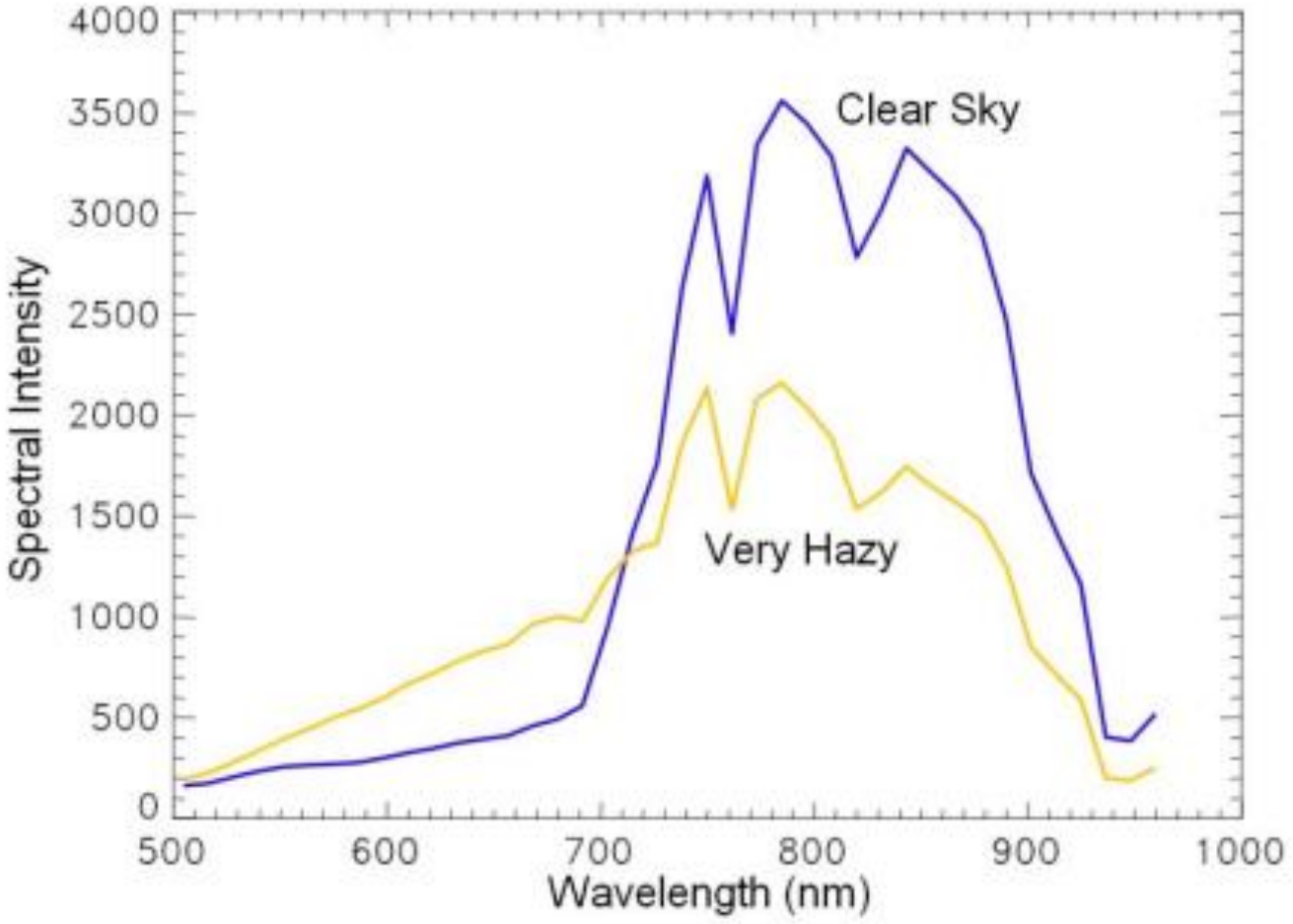
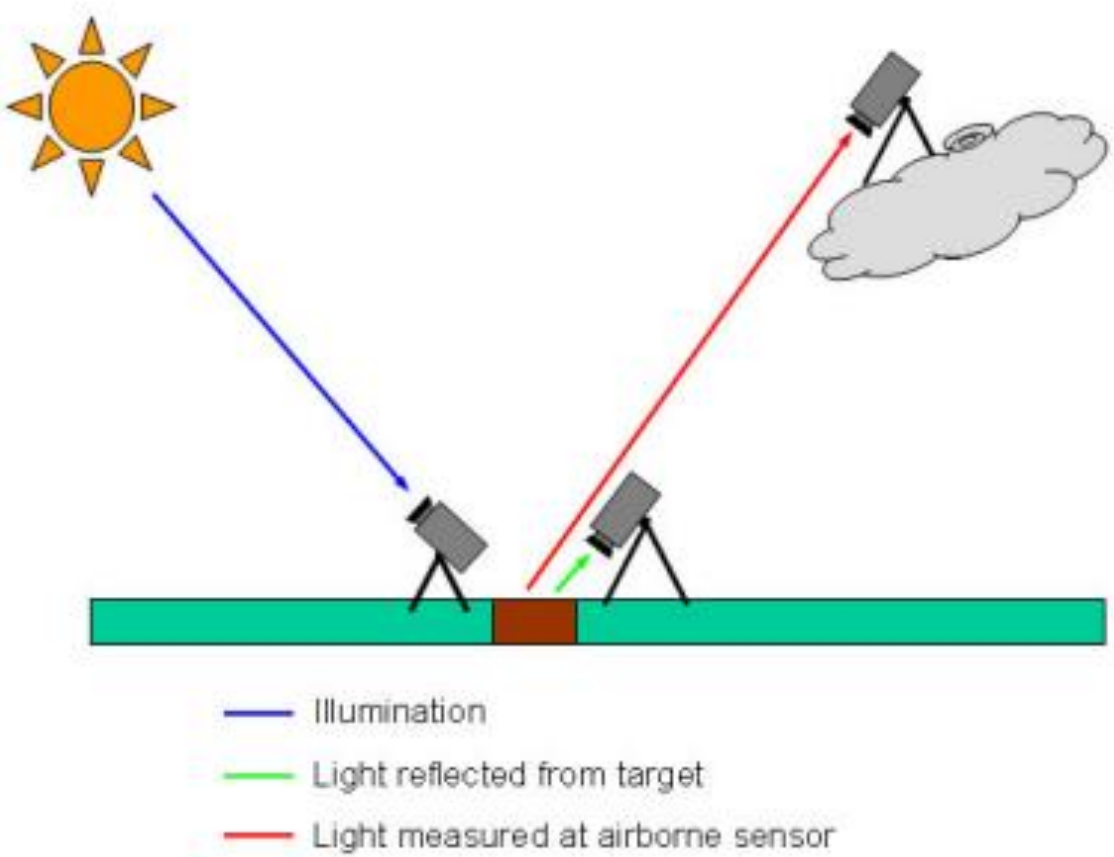
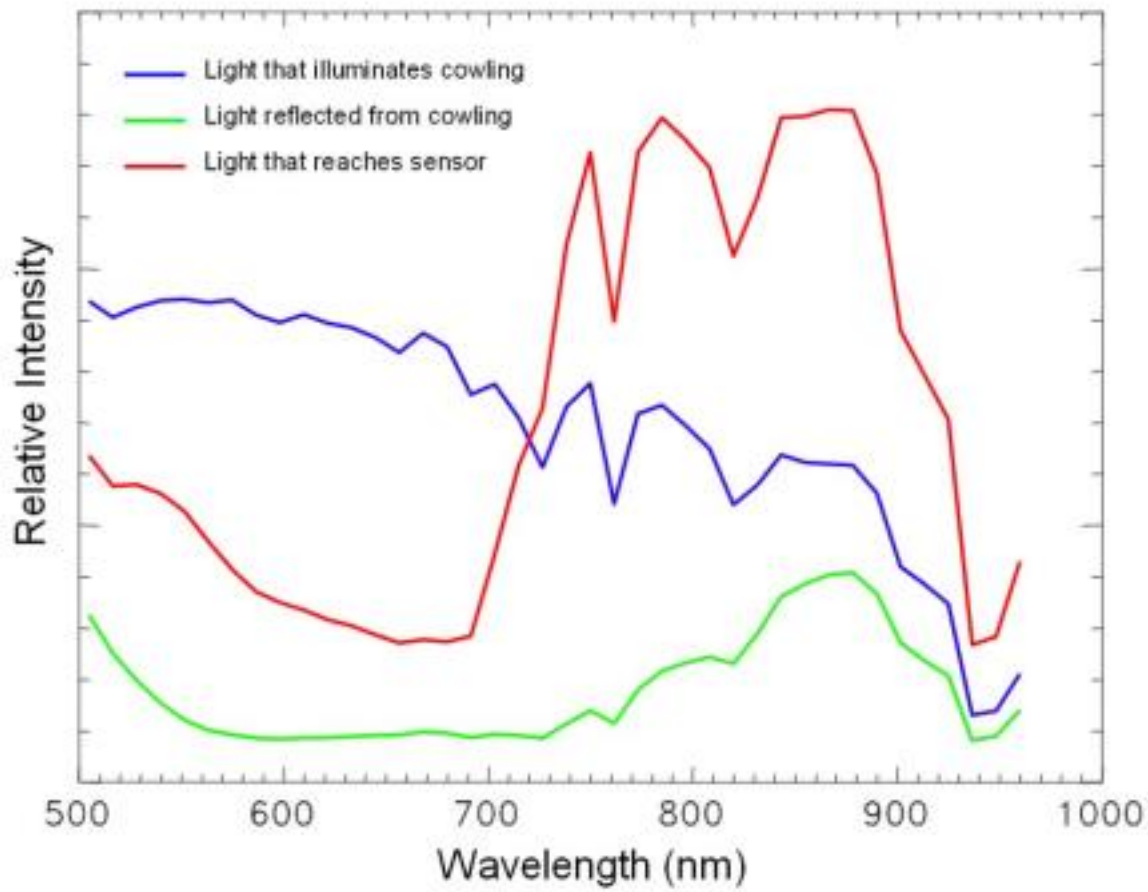


Figure 26. Spectral Intensities for Different Sky Conditions Blue Cowling Surrounded by Healthy Vegetation; Mid-Day Summer


Illumination
Light reflected from target
Light measured at airborne sensor


Light that illuminates cowling
Light reflected from cowling
Light that reaches sensor
Relative Intensity
500
600
700
800
900
1000
Wavelength (nm)

Geometric distortions

Acquiring images with airplanes are affected by several factors:

Platform motion and attitude

- plane vibrations, wind gusts, and sudden movements by the pilot.

Terrain relief and topography

- Rugged and undulated terrain, especially in mountaineous areas
  - Leads to perspective effects
    - Size and shape can appear different than in reality

Orthorectification errors

- Errors in elevation model cause misalignment between image and terrain model

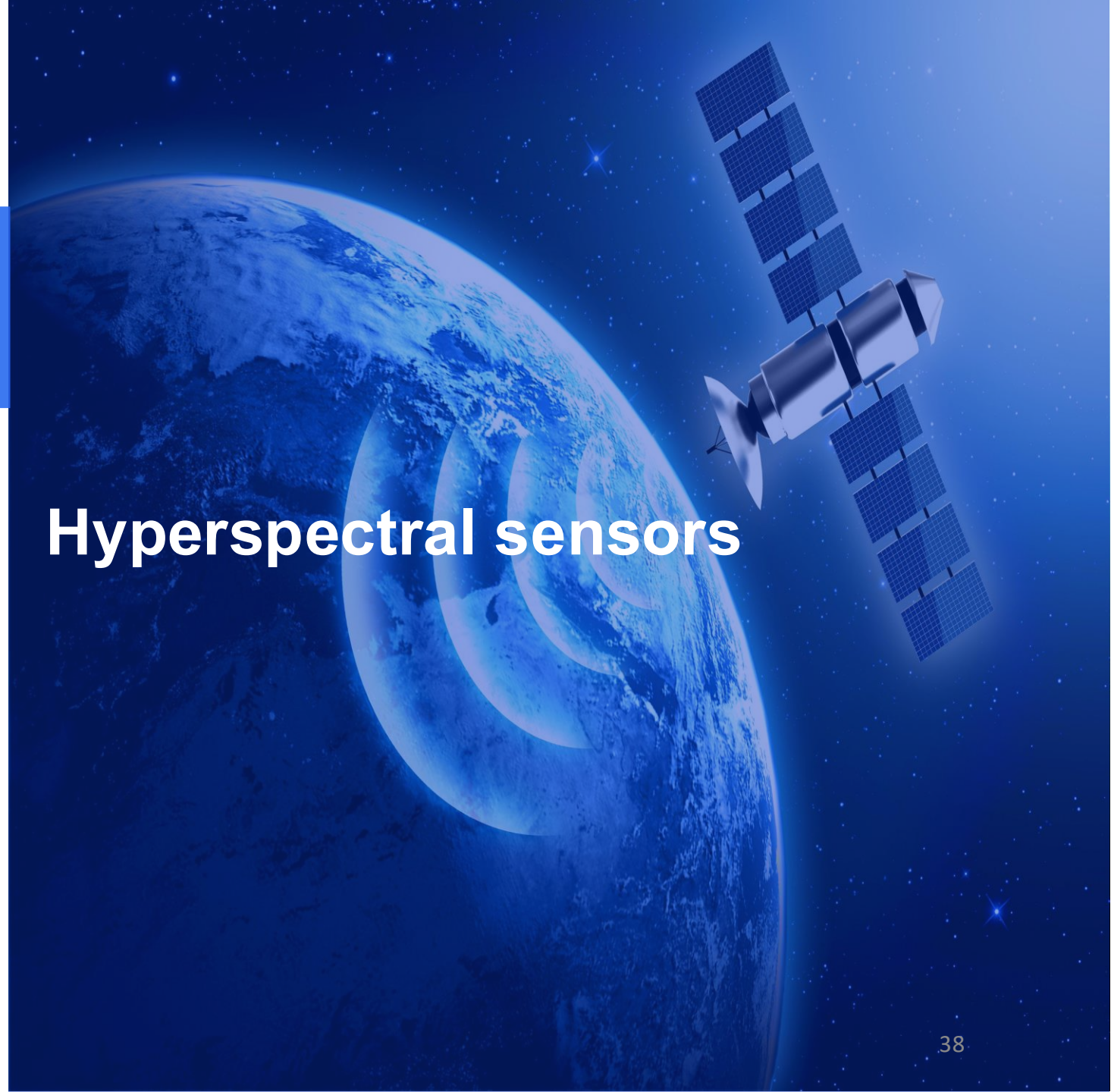


Challenges (2)

- Geometric distortions
  - Vibrations cause geometric shifts in the data
  - Sensor and IMU GPS mismatch
- Target conditions
  - Does the target's state represent what you want assess?
    - Seasonal variation in vegetation (leaf off vs leaf on)
    - Wet vs dry season
    - Early crop vs late crop cycle

6

Hyperspectral sensors



Hyperspectral satellite missions

Instrument	TianGong-1	PRISMA	HISUI	EnMAP	SHALOM	HypXIM
Organization	Chinese Academy of Science and Physics	Italian Space Agency (ASI)	Japanese ministry of Economy, Trade, and Industry	German GFZ-DLR	Italy-Israel Space agencies (ASI-ISA)	France Space Agency (CNES)
Date Range	2011-2013	2020-Present	2021	2021	2022	2021/2022
Spectral Range	400-2500 nm	400-2500 nm	400-2500 nm	420-2450 nm	400-2500 nm	400-2500 nm
Spectral Bands	128	249	185	244	275	210
Spatial Resolution	10-20 m	30 m	30 m	30 m	10 m	10 m
Objective	Land imaging in China	Natural resources and atmosphere	Energy, vegetation monitoring	Earth observation	Land and ocean observation	Soil, urban, coastal applications

Launch date	Name	Provider	Platform	Constellation size now / planned	Wavelengths in nm	Spectral res. in nm	Spatial res. in m
N/A	AVIRIS (classic)	NASA/JPL	Airborne		380-2510	10	Varies
N/A	AVIRIS-NG	NASA/JPL	Airborne		380-2510	5	Varies
2018	DESIS	DLR	ISS		400-1000	2.55	30
2019	PRISMA	ASI	Satellite		400-2500	10	30
2022	EnMAP (VNIR)	DLR	Satellite		420-1000	6.5	30
2022	EnMAP (SWIR)	DLR	Satellite		900-2450	10	30
2020-25	Satellogic Aleph-1	Satellogic	Satellite	38 / 200+	460-830	14-35	25
2022-23	PIXXEL	PIXXEL	Satellite	6			5
2023-24	Orbital Sidekick GHOSt	Orbital Sidekick	Satellite	3 / 6	400-2500	3-5	8.3
2023	Esper	Esper	Satellite	18	400-2000	6	6
2023	Carbon Mapper (Tanager)	Planet/JPL/CarbonMapper Coalition	Satellite	2		5	30
2023	Wyvern	Wyvern Space	Satellite	36			<5
2023	HySpec	HyspeqIQ	Satellite	12		105	5
2023	HyperSat	HyperSat	Satellite				6
2023	Kuva Hyperfield-1	Kuva Space	Satellite		450-1000		25
early 2024	PACE	NASA	Satellite	1	350-885	1.25 or 2.5	
2024	PIXXEL	PIXXEL	Satellite	30	400-2500		5
2027/28	SBG	NASA	Satellite			10	30
2029	CHIME (Sentinel 10)	ESA	Satellite		400-2500	10	
Unknown	BlackSky	BlackSky	Satellite	Unknown			

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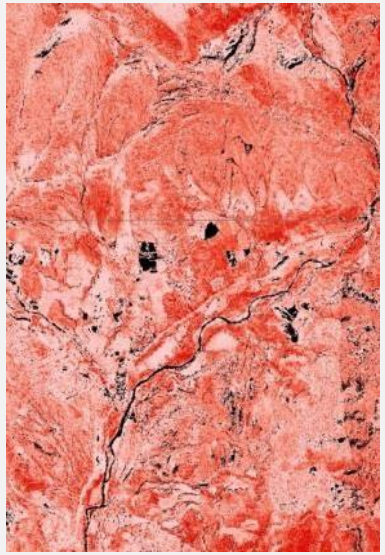
Applications

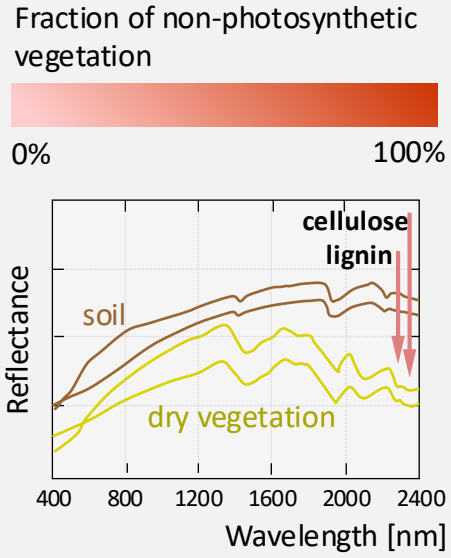


Applications environmental

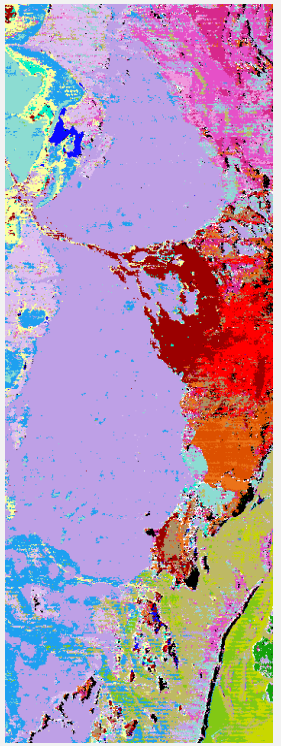
- Mine exploration
- Soil mineralogy, and texture
- Vegetation monitoring: plant stress, crop residues, plant composition

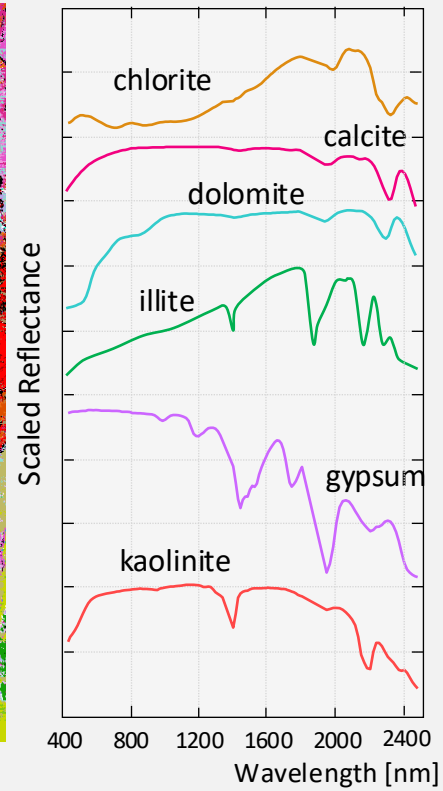
Dry vegetation Mapping





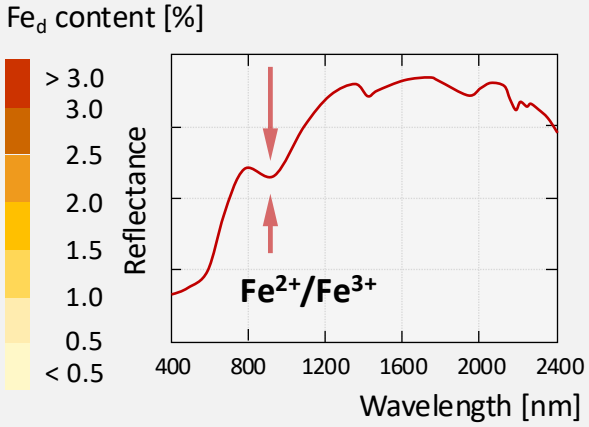
Mineral mapping





Soil mapping

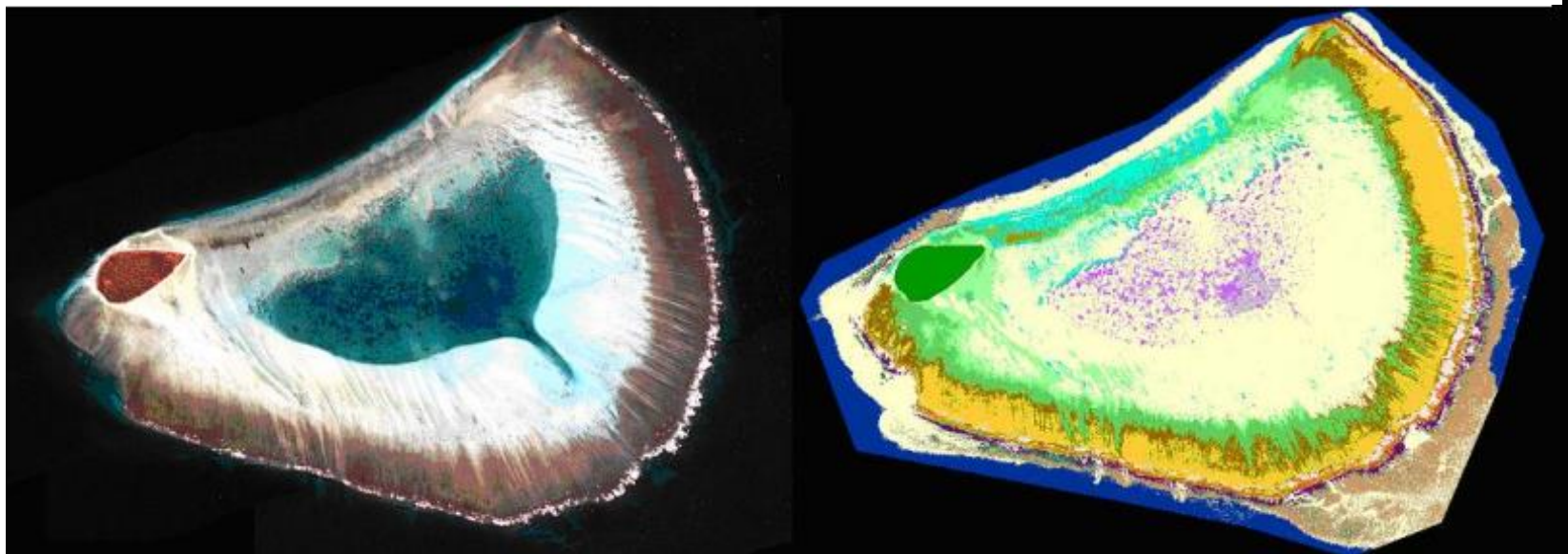


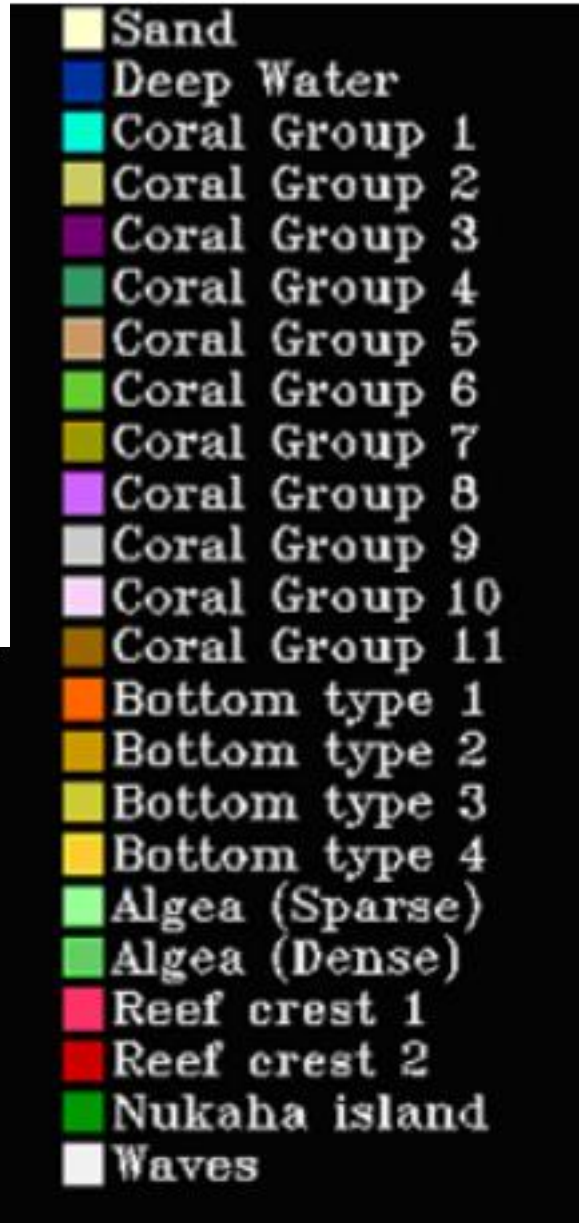


Applications environmental

- Geology
- Invasive Species
- Coastal/Ocean monitoring
- Carbon monitoring
- Microbial life in the Arctic
- Volcanic Activity

Left: The island of Fordate in Tanimbar (Indonesia) by CASI hyperspectral data (resolution 2,5 m). Right: The bottomtype classification map made from the hyperspectral data. Image Credit: SEOS

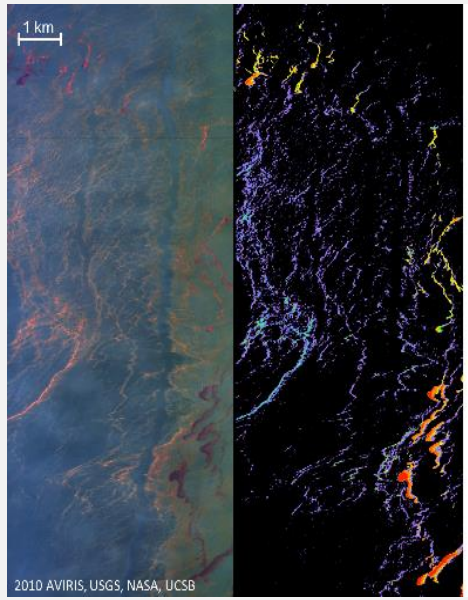


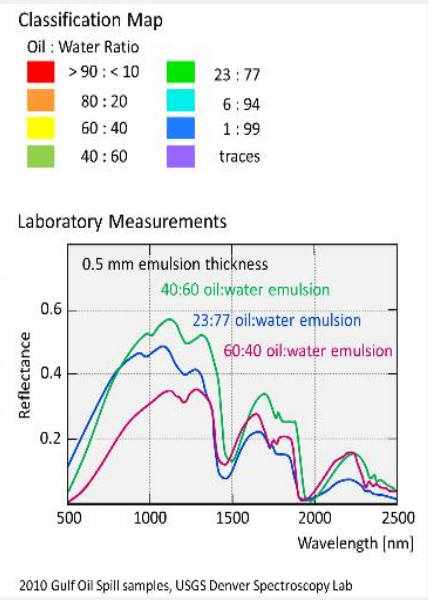


Applications environmental

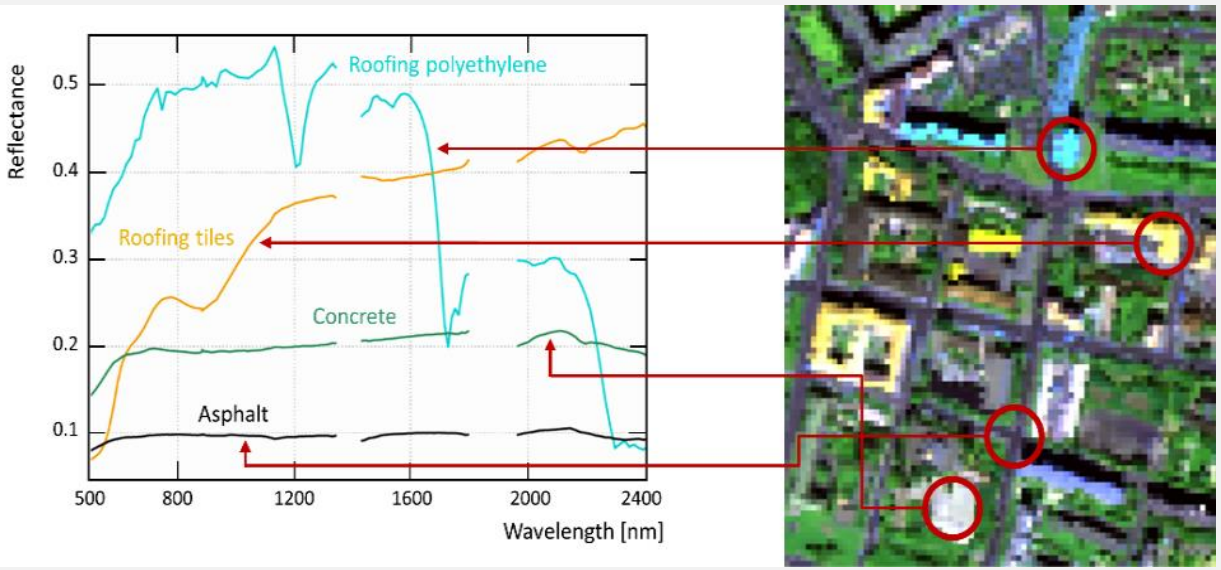
- Hazardous materials: oil, dust, mine waste, pollutants, plastics
- Inland and coastal waters: phytoplankton composition, (harmful) algae species
- Urban surface materials

POLLUTION MAPPING

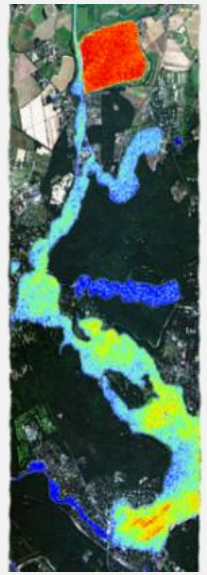


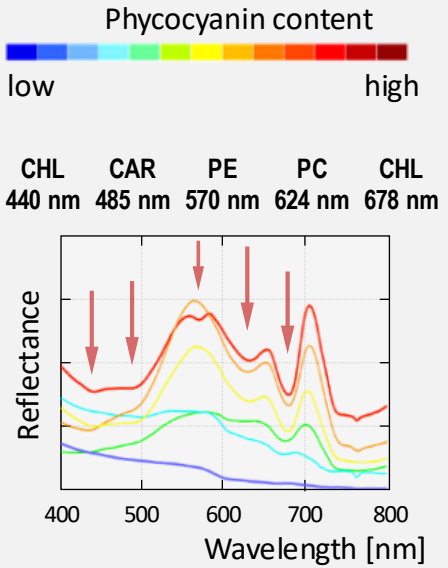


URBAN SURFACE MAPPING



WATER QUALITY MAPPING



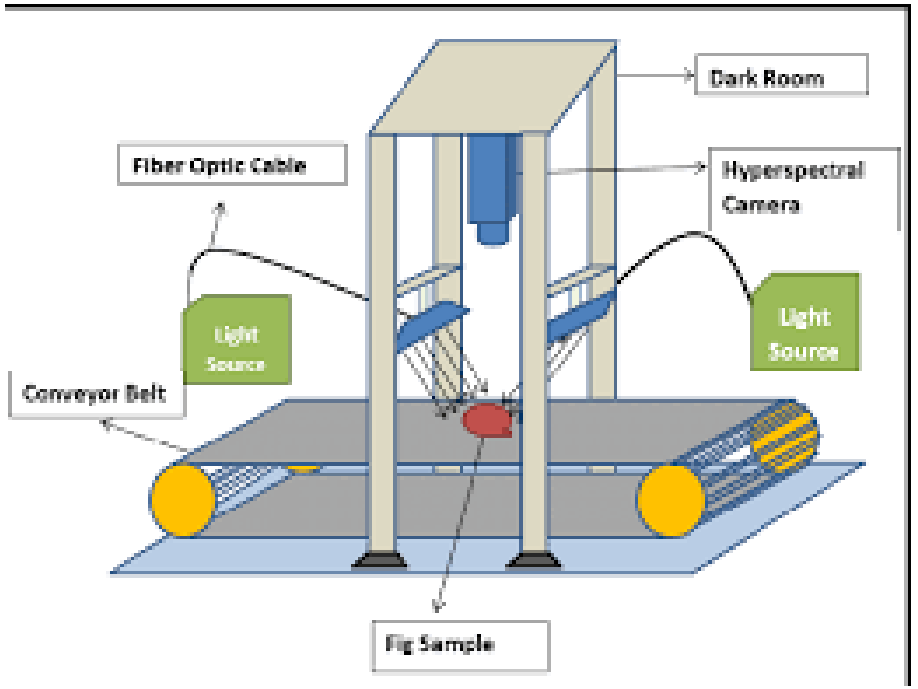


Applications manufacturing

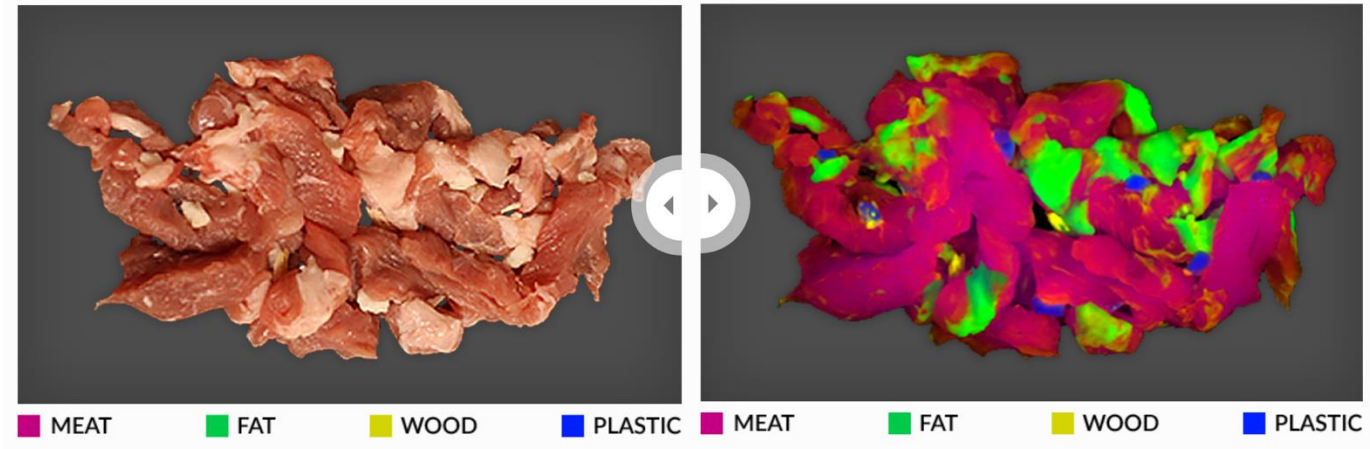
HSI is ideal for separating chemically different materials

Typical wavelengths used: 900-1700 nm region

Uses artificial halogen lamp sources



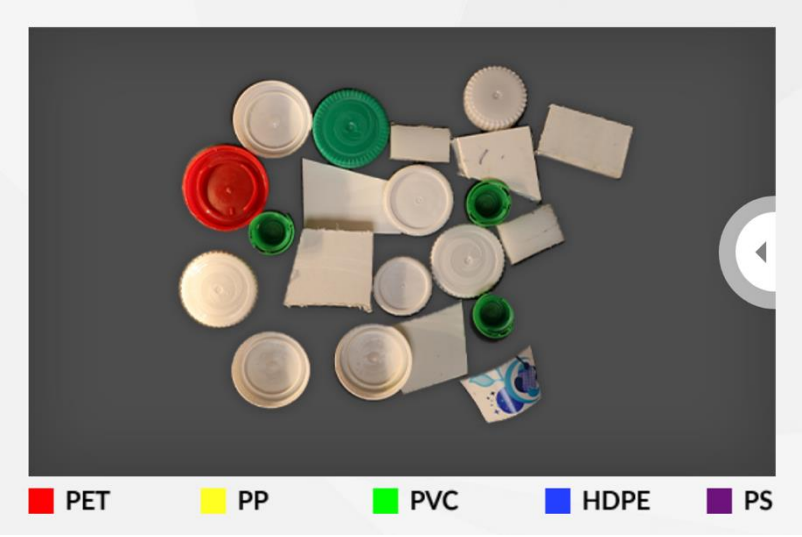
- Food industry

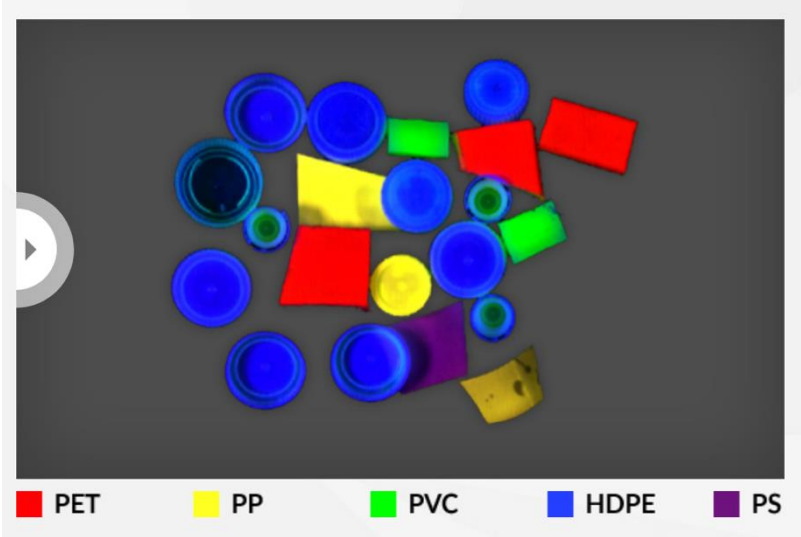




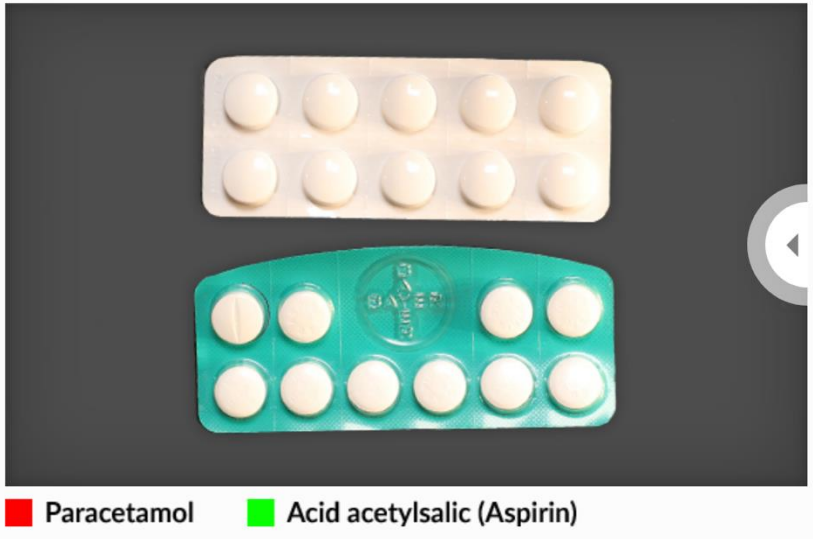
Applications

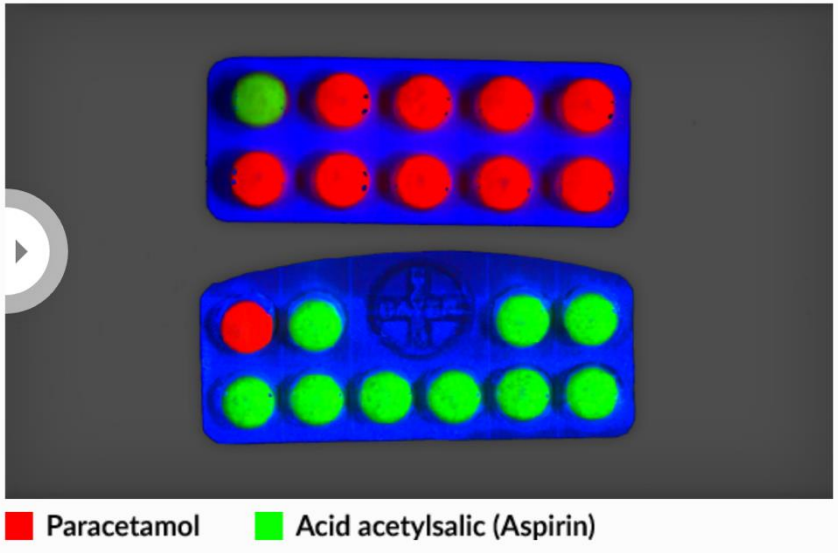
- Recycling & Waste management
  - Detect ‘alien’ objects



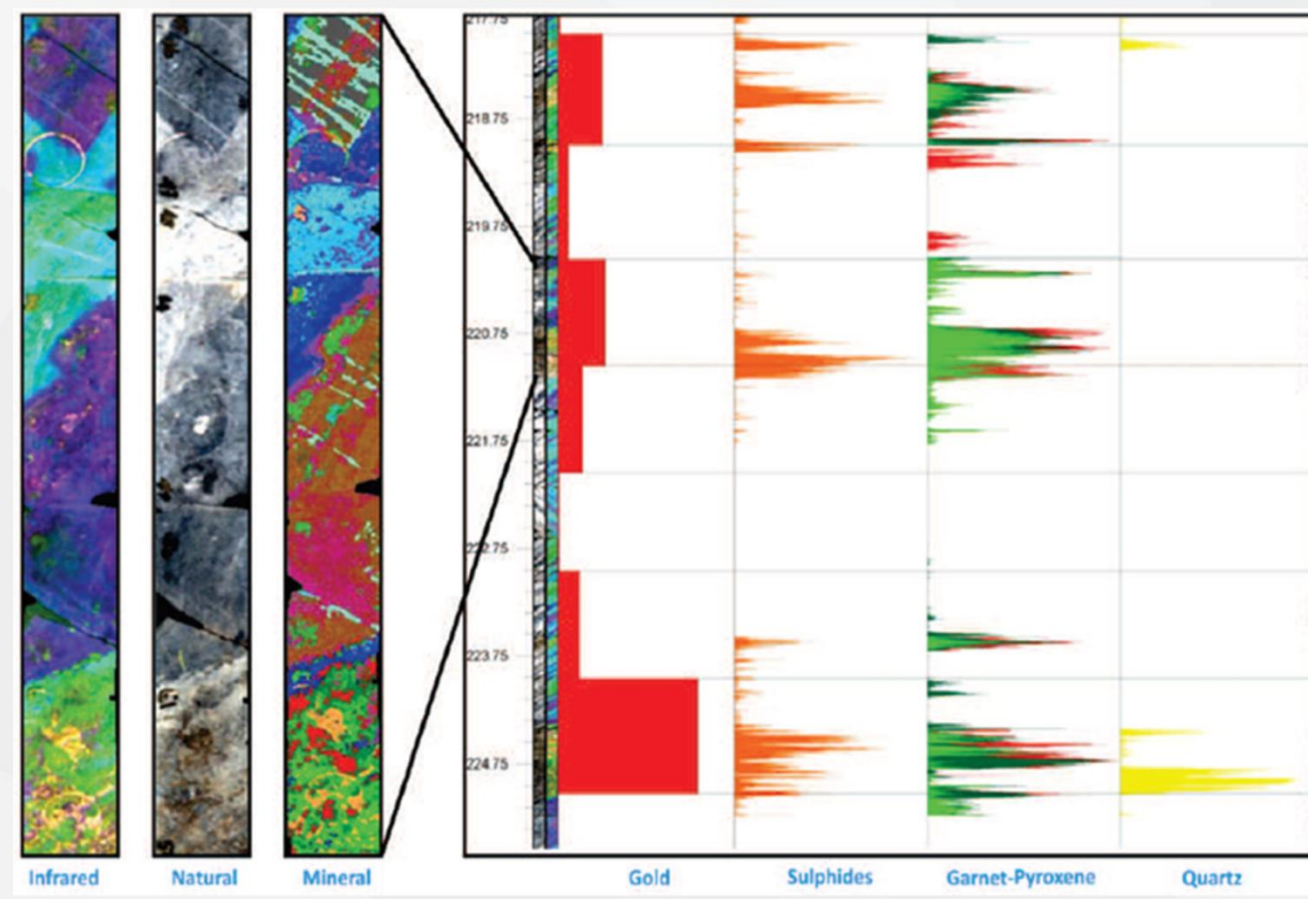


Applications manufacturing





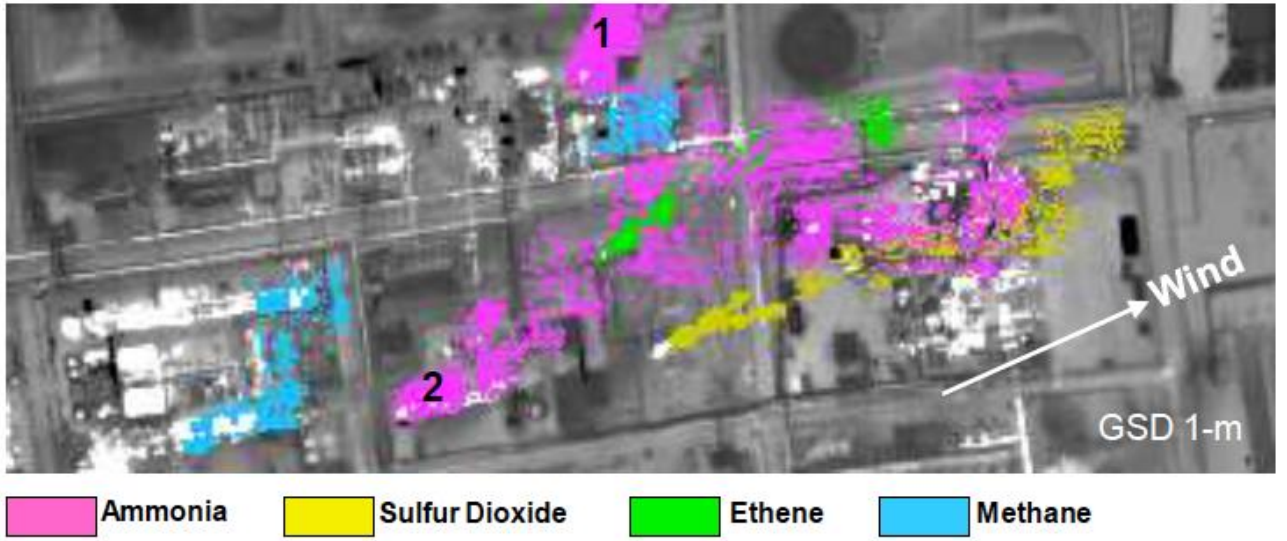
Pharmaceutical industry & Geological



Methane Detection

- Currently, Ground-based/Aerial systems using MWIR and LWIR multispectral systems can detect and monitor methane and other gases at high resolution for leaks detection.

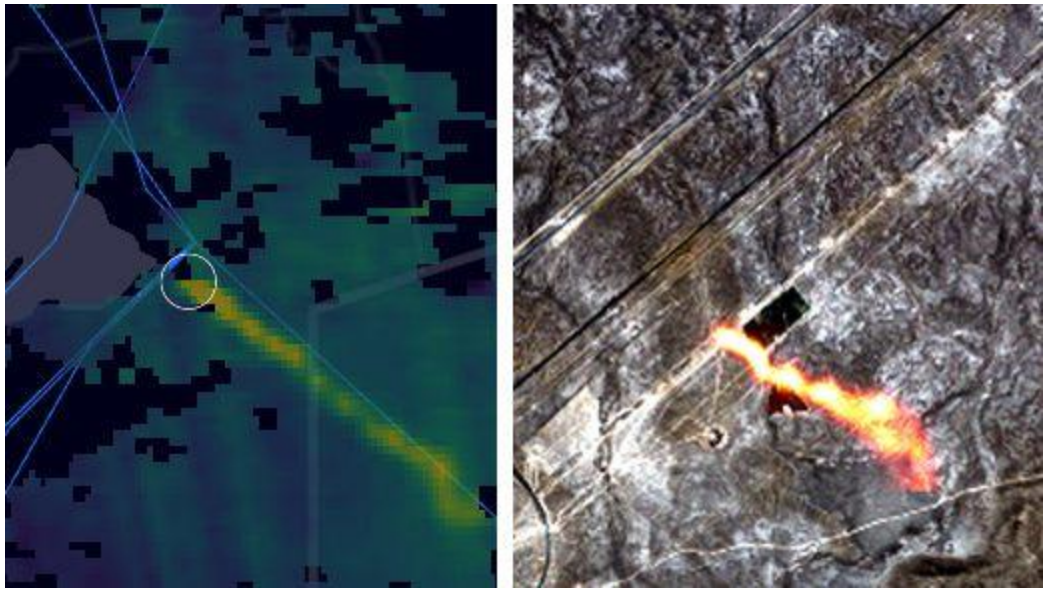
- In the near future, satellite constellations like GHGSat, Bluefield, etc. will provide continuous high resolution monitoring of emissions at 10-20 m GSD.



Source: Telops



Source: Telops



Example from Sentinel 2 and Sentinel 5P on Methane Pipeline lek in Kazhakstan.
Source: Copernicus

Current & future trends

- Data fusion
  - Lidar & HSI, or HSI & thermal



Hyperspectral satellites

1) Esa Flex mission

Launch: Dec. 2025

Focus: vegetation analysis

550 – 880 nm

Cube sats

On demand acquisitions

Pixxel, Esper, Planet, kuva space, Jilin-1

UAV/Drones

Sensors getting smaller and lighter

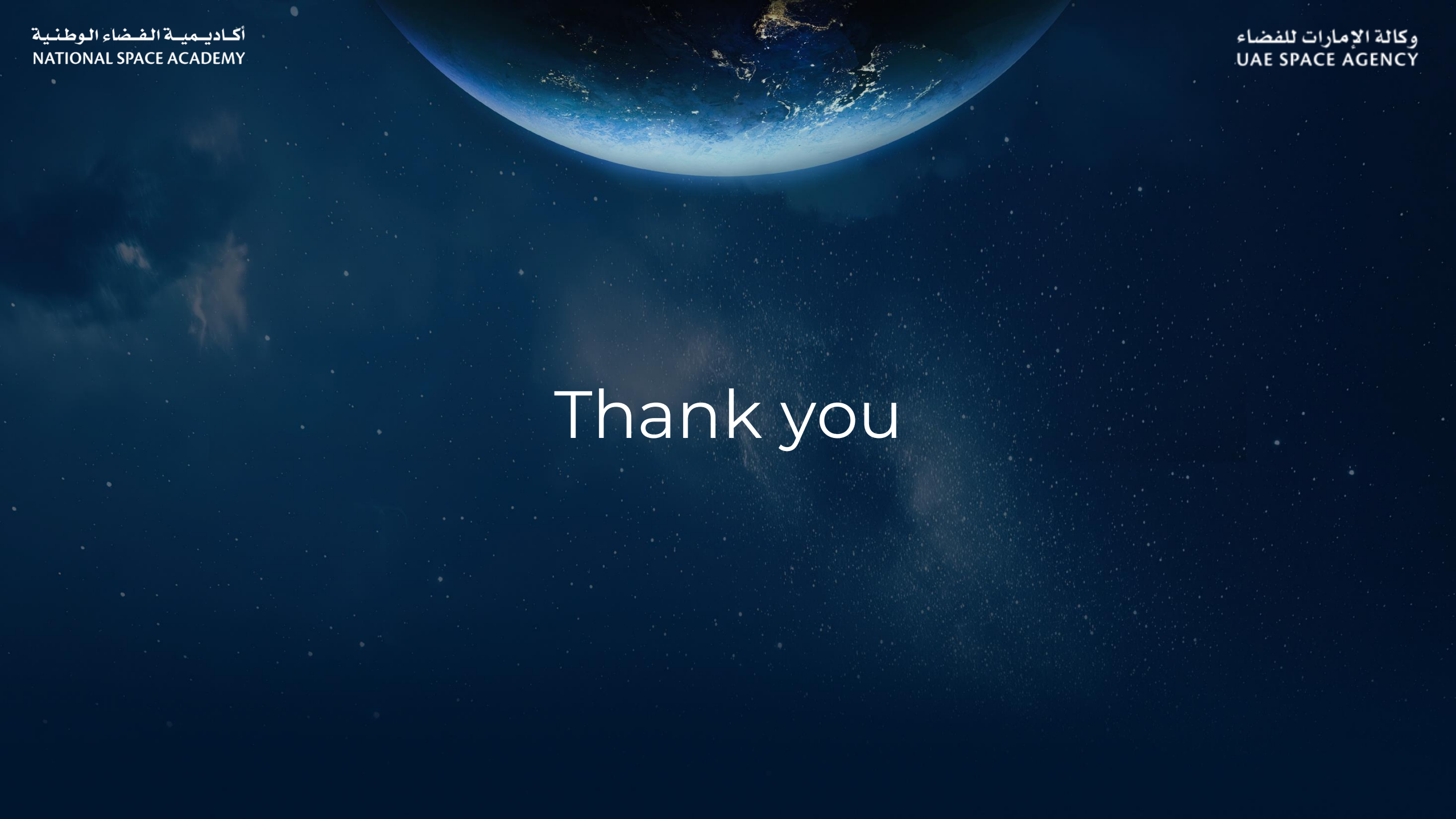


Summary

- HSI offers great benefits
- HSI data is large and complex
- HSI acquisition & processing does involve many challenges
- Powerful tool for detection and classification of similarly looking objects
- Future trends: satellites, cube sats and UAVs.
- HSI will become more affordable.

أكاديمية الفضاء الوطنية
NATIONAL SPACE ACADEMY

وكالة الإمارات للفضاء
UAE SPACE AGENCY



Thank you